What LLM Mathematics Benchmarks Measure: Construct Validity, Artifact Projection, and Certified Semantic Transport

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Abstract

Mathematics benchmarks for large language models increasingly report performance as evidence about mathematical reasoning ability or capability. This paper argues that the inference is not presently identified. MathArena, MathConstruct, the Open Proof Corpus, BrokenMath, and ProofRank provide valuable protocol-relative measurements: success on fresh competition problems, verifier-accepted constructions, grader-accepted proofs, rejection of corrupted claims, and selected proof-quality proxies. A score certifies that an evaluator accepted an artifact produced under a specified protocol; it does not by itself identify a representation-independent latent trait of mathematical reasoning. We formalize the gap by modelling a benchmark as an instrument acting through an artifact projection, writing the score-to-capability step as an identification problem, and proving elementary non-identification results for artifact-only and finite-sample evaluation. We then develop certified semantic transport: presentation-indexed answer spaces, a groupoid of invertible certified equivalences, an answer functor into sets and bijections, and an equivariant correctness subfunctor in **Set**. This yields one additional validity test—whether correctness commutes across certified reformulations—without claiming that invariance exhausts mathematical reasoning. An instrument-by-instrument audit states what each measures well and which broader interpretation is unsupported. Mathematical physics supplies stress cases: gauge redundancy, local-to-global obstruction, anomalies, regularization, boundaries, and duality as a target not yet instantiated. No model result is asserted: the contribution is a critique of score interpretation, a corrected measurement framework, and a pre-specified design whose measurement core, analysis prototype and calibration bank ship as ancillary files.

Keywords: large language models; mathematical reasoning; construct validity; measurement models; identification; equivariance; groupoids; gauge invariance; anomalies; benchmark reporting.

Contents

1 Introduction: the score-to-capability inference **3**

1.1 A score is not self-interpreting 3

1.2 The thesis, stated exactly 4

1.3 What this paper does 4

1.4 Scope, and what is not claimed 5

2 The claim and operationalization ledger **5**
2.1 Capability language, in context . . . 5
2.2 What is directly observed . . . 6

3 Benchmarks as Measurement Instruments **7**
3.1 The instrument as a six-tuple . . . 7
3.2 The identification problem . . . 7
3.3 Reliability is not validity . . . 8

4 Cross-Cutting Construct-Validity Limits **9**
4.1 Artifact projection and collapse . . . 9
4.2 Finite-sample non-identification . . . 9
4.3 Verifier-selection of the task distribution . . . 10
4.4 Judge substitution . . . 10
4.5 Aggregation and dimensionality . . . 11
4.6 Formalization adequacy . . . 11

5 Instrument-by-Instrument Audit **12**
5.1 MathArena . . . 12
5.2 The MathArena platform . . . 13
5.3 MathConstruct . . . 13
5.4 The Open Proof Corpus . . . 14
5.5 BrokenMath . . . 14
5.6 ProofRank . . . 15
5.7 What the audit shows . . . 15

6 Certified Semantic Transport **16**
6.1 Presentations, certified equivalences, and what is excluded . . . 16
6.2 The answer functor and the correctness subfunctor . . . 17
6.3 Normalization, quotient validity, and the equivariance condition . . . 19
6.4 The invariance defect . . . 21
6.5 Certificates and anti-certificates . . . 23
6.6 Relation to prior work on invariance and perturbation . . . 24

7 What IMR Measures—and What It Does Not **25**

8 Mathematical Physics as the Stress Domain **26**
8.1 Gauge redundancy: the observable is on the quotient . . . 26
8.2 Local-to-global obstruction . . . 27
8.3 Anomalies as measure Jacobians . . . 27
8.4 Regularization and limit interchange . . . 28
8.5 Boundary terms . . . 28
8.6 Duality—and what a pilot module can honestly claim . . . 29

9 A Minimal Empirical Design **29**

9.1 The family: the minimum viable experimental unit . . . 29
9.2 Generation protocol . . . 30
9.3 The obligation record . . . 32
9.4 Three separated estimands . . . 32
9.5 The confirmatory endpoint and its decision rule . . . 34
9.6 Analysis specification . . . 37
9.7 Attribution taxonomy and the evaluator gate . . . 40
9.8 Hypotheses and threats to validity . . . 41

10 Implications for Benchmark Reporting **43**
10.1 A claim matrix . . . 43
10.2 Six reporting items . . . 43

11 Conclusion **44**

A Notation **45**

B The Pilot Family Bank and Released Artifacts **45**

1 Introduction: the score-to-capability inference

1.1 A score is not self-interpreting

A benchmark score is not self-interpreting. It is the output of an instrument: a task distribution, a prompt and tool protocol, a response format, an evaluator, and a scoring rule. Over the last two years the instruments available for evaluating large language models (LLMs) in mathematics have improved substantially in freshness, breadth, verifiability, and inspectability. MathArena evaluates on competitions released after model cutoffs [1]; its research platform extends the programme to research-derived problems, corrupted claims, and formal statements [2]; MathConstruct grades constructed objects with executable verifiers [3]; the Open Proof Corpus collects thousands of human-graded natural-language proofs [4]; BrokenMath measures sycophantic acceptance of falsified theorems [5]; ProofRank scores declared proxies of proof quality [6]. These are real methodological advances, and this paper takes them as the state of the art.

Several of these works, and much of the surrounding discourse, describe what is measured using the language of mathematical *reasoning ability* or *capability*. The central question of this paper is whether the published observations identify that broader construct.

Our answer is negative in a precise and bounded sense. The instruments directly observe accepted mathematical artifacts produced under specified protocols. Their scores support claims about performance on those tasks under those protocols. Without an explicit measurement model, evidence of generalization to a declared target population, and invariance tests across equivalent presentations, the same scores do not identify a representation-independent latent trait of mathematical reasoning. This is a claim about the interpretation of scores. It is not a claim about the motives of benchmark authors, not a claim that the measured performances are illusory, and not a claim that models do not reason—a proposition this paper neither asserts nor is equipped to test.

1.2 The thesis, stated exactly

> MathArena, MathConstruct, the Open Proof Corpus, BrokenMath, and ProofRank provide scientifically useful measurements of model performance on selected mathematical tasks under specified protocols. Their scores do not, without additional construct-validity evidence, identify the broad, representation-independent latent construct described as mathematical reasoning ability or capability.

The thesis is sharper than "these benchmarks do not measure reasoning," and it is defensible where the cruder claim is not. It separates five statements that the discourse routinely conflates:

$$\begin{aligned} &\underbrace{\text{artifact acceptance}}_{\text{L0}} \neq \underbrace{\text{task performance}}_{\text{L1}} \neq \underbrace{\text{domain generalization}}_{\text{L2}} \\ &\neq \underbrace{\text{representation-stable correctness}}_{\text{L3}} \neq \underbrace{\text{broad mathematical capability}}_{\text{L4}}. \end{aligned} \tag{1}$$

The audited corpus supplies substantial evidence at L0–L1 and selected evidence toward L2. This paper's positive contribution, Invariant Mathematical Reasoning (IMR), targets L3. Neither the corpus nor IMR identifies L4. The five are written in a line for readability, but they are not a ladder in the logical sense: L3 does not imply L2, since transport stability on a finite certified orbit says nothing about a fresh problem drawn from the target population, and L2 does not imply L3. They are distinct evidence requirements, and the relation between them is a partial order at best. Section 10 turns (1) into a reporting discipline. The five are evidence dimensions, not rungs of a ladder.

1.3 What this paper does

(i) It writes the inference down. Section 3 models a benchmark as a six-tuple and states the score-to-capability step as an identification problem: the observed score is a function of a broad capability parameter Θ_{math}, of nuisance response processes η_M (memorization, search, tool use, formatting competence, refusal policy, protocol adaptation), and of the instrument itself. Reliability, contamination control, and grader agreement constrain the error term; none of them separates Θ_{math} from η_M.

(ii) It proves the elementary limits. Section 4 establishes three non-identification results—artifact-level, finite-sample, and quotient-level—and analyses three further cross-cutting limits: verifier-selection of the task distribution, judge substitution, and the adequacy gap between a formal statement and its informal source. The results are deliberately elementary. No impossibility theorem from logic (Rice, Tarski, Löb, Gödel, Chaitin) is load-bearing anywhere in this paper; the foundational literature enters only as bounded, correctly transferred consequences.

(iii) It audits instrument by instrument. Section 5 applies one fixed six-question template to each instrument: what capability language is used; what is directly observed; what evaluator and protocol produce the label; what inference the design licenses; which broader inference is not identified; and what additional test would close the gap. Each subsection concedes what the instrument establishes before stating what it does not.

(iv) It supplies one missing test, correctly formalized. Section 6 develops certified semantic transport: a groupoid $\mathscr{G}_{\text{IMR}}$ of *invertible* certified equivalences, an answer functor $A : \mathscr{G}_{\text{IMR}} \to \mathbf{Set}_{\cong}$, and an equivariant correctness subfunctor $C \hookrightarrow U \circ A$ taken in $\mathbf{Set}$, with correctness on a transformed presentation judged against transported acceptable answers. Section 7 states the symmetric limitation: transport-stable correctness is necessary evidence for representation independence, not a complete theory of mathematical reasoning.

(v) It proposes a minimal design and releases its instrument. Sections 8–9 develop mathematical physics as the adversarial domain and pre-specify a measurement protocol with separated estimands,

a declared missingness rule, a margin-based decision rule, a multiplicity policy, and an evaluator gate. Appendix B summarizes a twenty-four-family pilot bank; its reference implementation, materialized family artifacts, executable certificates, and test suite accompany this paper as ancillary files.

1.4 Scope, and what is not claimed

No model-performance result is reported in this paper, because none has been measured. Every statement about how models behave under certified transformation is a labelled hypothesis, and the labels are not uniform: H1 and H4 are decision-bearing, with pre-specified rules; H2 is descriptive and carries no test; H3 lies outside the released instrument and is a design target (Section 9); a null outcome—correctness stable across all validated transformations—is a possible, reportable, and theory-revising result to which the design commits in advance, and would constitute evidence that existing scores generalize across certified equivalence better than this analysis anticipates.

Three further restrictions bind the argument. First, the critique targets score interpretation, never authorship: it is designed to remain correct after all names and institutional affiliations are removed. Second, every concession that an instrument's own authors state is recorded before the corresponding criticism, and criticism of a stated limitation is never presented as discovery of a hidden one. Third, the same restraint applies reflexively to IMR: it measures answer-level and obligation-level transport stability, never latent understanding, and it does not reach conjecture formation, definition choice, explanatory value, or discovery.

2 The claim and operationalization ledger

The audit begins with evidence rather than characterization. This section records, for each instrument, the capability language used, and then the random variable actually observed. Quotations are short and given in context; where a paper states its own limits, the limitation is recorded here and not later re-presented as a finding.

2.1 Capability language, in context

MathConstruct is titled as a challenge to LLM reasoning through constructive proofs, and describes the benchmark as designed to [3, p. 2] "evaluate LLMs' reasoning capabilities through constructive proofs"; its opening sentence frames the work as the evaluation of mathematical reasoning abilities [3, p. 1]. *MathArena* describes its aim as a [1, p. 2] "rigorous and up-to-date evaluation of mathematical reasoning", and—this is a point in its favour, and the reason its design is strong—explicitly identifies the inferential problem it solves: with widely available evaluation sets it is difficult to separate genuine reasoning from memorization, which is precisely why the benchmark evaluates only on competitions released after model cutoffs [1, p. 1]. The *MathArena platform* describes the need for [2, p. 1] "accurate, up-to-date evaluations of mathematical capability"; in the same paragraph it states that no single benchmark spans the full range of mathematical capabilities, and its discussion section records that problem-solving performance omits other important mathematical capabilities such as interaction, conjecturing, and communication [2, pp. 1, 9]. The *Open Proof Corpus* situates contamination control as a challenge for [4, p. 18] "evaluating the true reasoning capabilities of models", and independently notes that final-answer benchmarks cannot span mathematical capabilities because they never require a proof [4, p. 1]. *BrokenMath* makes the narrowest claim of the six: it presents itself as a benchmark for sycophancy in theorem proving, and its stated target is the behavioural phenomenon, not general proving ability [5, p. 1]. *ProofRank* is the most careful: it introduces its five dimensions explicitly as [6, p. 1] "scalable proxies of proof quality", a qualification this paper takes at face value and does not argue against.

Instrument	Directly observed random variable	Licensed inference	Not identified by the design
MathArena [1]	Correctness of final answers, and grader-scored proofs, on competitions released after model cutoffs	Lower-contamination task performance on those competitions	Stability under equivalent reformulation; freshness is a temporal property, not a semantic one
MathArena platform [2]	Heterogeneous tiers—final answers, proofs, corrupted-claim rejection, formal statements—plus a fitted 2PL item-response summary	Protocol-specific performances, and a rank-stable expected-performance index	That the fitted scalar θ_m is a broad capability: unidimensionality, local independence, cross-tier invariance, DIF, and the imputation mechanism are untested
MathConstruct [3]	Executable-verifier acceptance of a constructed witness, plus success across sampled parameter variations	Constructive-task performance and robustness on the sampled variants	The response process that produced the witness; uniformity beyond the sampled parameters
Open Proof Corpus [4]	Human (and LLM) correctness judgments on natural-language proof artifacts	Proof performance under a specified rubric; the answer/proof discrepancy	Judge accuracy from agreement alone; proof competence outside the competition distribution
BrokenMath [5]	Whether a model resists or endorses expert-refined false statements	Sycophantic response behaviour on that corruption distribution	Robustness to structural obstructions absent from the corruption process
ProofRank [6]	Judge assessments on five declared proxy dimensions	Differences on the declared proxies, as the authors state	A foundation-independent ordering of proof quality, insight, or originality
Formal tier [2, 15]	Kernel acceptance of a term against a formalized statement in a fixed library	Formal correctness relative to that statement and environment	Adequacy of the formal statement to its informal source (Proposition 4.5)

Table 1: The operationalization ledger. The final column is the subject of this paper: in each row it names *a possible broader inference that the design, as published, does not license*. It is not a claim that each instrument's authors draw it—BrokenMath scopes itself to a behaviour and ProofRank labels its dimensions as proxies—but that the inference is available to a reader of the score and is not warranted by the design.

Two observations follow immediately, and they set the tone for everything after. First, the capability language is real, recurrent, and consequential—it is how these numbers are read by the wider community. Second, the corpus is not uniform: the platform and the OPC state coverage limits, ProofRank labels its constructs as proxies, and BrokenMath scopes itself to a behaviour. A critique that treats the six instruments as making one undifferentiated overclaim would be inaccurate. The argument of this paper is therefore not that the authors are unaware of the gap, but that the gap is not closed by the designs as they stand, and that the missing evidence is specifiable and collectible.

2.2 What is directly observed

Table 1 maps each instrument to the variable it actually records. The pattern across rows is uniform and is the structural claim of the paper: what is observed is always *acceptance of an artifact by a specified evaluator under a specified protocol*. Every additional step—from acceptance to task performance, from task performance to a target population, from performance on canonical presentations to correctness under equivalent ones, and from any of these to a capability parameter—is an inference requiring its

own warrant. Sections 3 and 4 supply the formal apparatus for saying exactly which warrant is missing where.

3 Benchmarks as Measurement Instruments

3.1 The instrument as a six-tuple

Definition 3.1 (Benchmark instrument)**.** A benchmark is the six-tuple

$$\mathscr{B} = \big(D_{\mathscr{B}},\, A_{\mathscr{B}},\, J_{\mathscr{B}},\, h_{\mathscr{B}},\, \Gamma_{\mathscr{B}},\, R_{\Gamma_{\mathscr{B}}}\big), \tag{2}$$

where $D_{\mathscr{B}}$ is the distribution over problem records; $A_{\mathscr{B}}$ the admissible artifact space (an integer, a symbolic expression, a matrix or graph, a proof text, a formal term, a behavioural label); $J_{\mathscr{B}}$ the judge kernel (exact-answer parser, executable verifier, proof-assistant kernel, human panel, LLM judge, or ensemble); $h_{\mathscr{B}}$ the scoring map that converts judged records into a reported number; $\Gamma_{\mathscr{B}}$ the protocol (system prompt, sampling temperature, token budget, attempts, tool access, verifier feedback, selection among candidates, post-processing, grading instructions); and $R_{\Gamma_{\mathscr{B}}}$ the prompt renderer that turns a record into the text the model actually receives. Write $\mathscr{S}_n = \{(x_i, y_i, J_{\mathscr{B}}(x_i, y_i))\}_{i=1}^{n}$ for the judged sample obtained by drawing records from $D_{\mathscr{B}}$, rendering them through $R_{\Gamma_{\mathscr{B}}}$, sampling responses under $\Gamma_{\mathscr{B}}$, and judging. The reported score is the *sample statistic*

$$\widehat{S}_{\mathscr{B}}(M) = h_{\mathscr{B}}\big(\mathscr{S}_n\big), \qquad \text{with population target} \qquad S_{\mathscr{B}}(M) = h_{\mathscr{B}}\big(P_{X,Y,J}\big), \tag{3}$$

two different objects: $\widehat{S}_{\mathscr{B}}$ is a random variable computed from n judged records, and $S_{\mathscr{B}}$ is the corresponding functional of the law those records are drawn from. Sampling variability separates the two, and nothing in this paper's argument turns on that separation: the criticism concerns what the target functional $S_{\mathscr{B}}$ can identify, not how well the statistic $\widehat{S}_{\mathscr{B}}$ estimates it. The scoring functional $h_{\mathscr{B}}$ is general enough to cover the instruments actually in use: plain means, best-of-n selection, missing-cell imputation, latent-variable summaries such as (9), rank constructions, and vector-valued rubrics. The familiar special case is the double expectation

$$S_{\mathscr{B}}(M) = \mathbb{E}_{x\sim D_{\mathscr{B}}}\, \mathbb{E}_{y\sim M_{\Gamma_{\mathscr{B}}}(\cdot\,|\,R_{\Gamma_{\mathscr{B}}}(x))}\big[J_{\mathscr{B}}(x,y)\big], \tag{4}$$

recovered when $h_{\mathscr{B}}$ is the sample mean.

Exposing $h_{\mathscr{B}}$ and $R_{\Gamma_{\mathscr{B}}}$ as first-class slots is not pedantry. It is what makes attribution possible: a measured difference between two systems can originate in the model, in the judge, in the scoring map, or in the rendering of the problem, and an instrument that hides two of those four slots cannot separate them. The quantity evaluated by (3) is not a model but a compound system, $(\text{model}, \text{renderer}, \text{protocol}, \text{tools}, \text{sampling}, \text{judge}, \text{scoring}, \text{data})$. Two leaderboards may therefore rank the same models differently without either being wrong: they are measuring different compound systems.

3.2 The identification problem

Write the observation as a structural equation rather than as a direct reading of a trait:

$$\boxed{Y_{\mathscr{B}}(M) \;=\; \Phi_{\mathscr{B}}\big(\Theta_{\text{math}}(M),\; \eta_M;\; D_{\mathscr{B}}, \Gamma_{\mathscr{B}}, R_{\Gamma_{\mathscr{B}}}, J_{\mathscr{B}}, h_{\mathscr{B}}\big) \;+\; \varepsilon.} \tag{5}$$

Here $\Theta_{\text{math}}(M)$ is the broad capability parameter that capability language invokes, and η_M collects the alternative response processes that can produce an accepted artifact: retrieval of a memorized solution,

template completion, guess-and-check against a verifier, tool-mediated enumeration, formatting competence, refusal policy, and adaptation to the protocol itself. ε is measurement noise, including judge noise—whose rate is *not* identified by inter-rater agreement alone, since that identification requires a model of correlated grader error and of item prevalence [4, 31].

Equation (5) is a *schematic decomposition*, not a theorem: it names the parameters whose separation an interpretation requires, and it is useful precisely because it makes the requirement statable. The requirement is the usual one. Say two parameter values are *observationally equivalent* for $\mathscr{B}$ when they induce the same law of the judged sample:

$$(\Theta_{\text{math}}, \eta) \sim_{\mathscr{B}} (\Theta'_{\text{math}}, \eta') \iff P_{\mathscr{S}_n}\big(\Theta_{\text{math}}, \eta\big) = P_{\mathscr{S}_n}\big(\Theta'_{\text{math}}, \eta'\big). \tag{6}$$

Θ_{math} is identified by $\mathscr{B}$ only if $\sim_{\mathscr{B}}$ implies $\Theta_{\text{math}} = \Theta'_{\text{math}}$. Two objects must be kept apart throughout: the *reported statistic* $\widehat{S}_{\mathscr{B}}(M) = h_{\mathscr{B}}(\mathscr{S}_n)$, a function of a finite realized sample, and its *target functional* $S_{\mathscr{B}}(M) = h_{\mathscr{B}}(P_{X,Y,J})$, the corresponding functional of the population law. Sampling variability separates the first from the second; identification concerns whether the second determines Θ_{math}. A benchmark can estimate its target functional perfectly and still identify nothing about Θ_{math}, and this paper's criticism is about the second step throughout, never about sampling error. With $\Phi_{\mathscr{B}}$ unrestricted this fails trivially—a change in Θ_{math} can be absorbed by a compensating change in η—so identification is not a property one can hope to read off a benchmark; it is a property of the restrictions one is willing to impose and defend. The rest of this section asks which restrictions the audited designs actually supply.

Remark 3.2 (What the corpus establishes, and what (5) withholds)**.** Contamination control shrinks one component of η_M on one axis: freshness makes verbatim recall of the specific released problems implausible [1]. Executable verification makes acceptance deterministic and removes human and LLM grading variance from the correctness channel for constructed objects [3]—conditional on the verifier being correct and its specification adequate, since a buggy, incomplete, or exploitable verifier introduces error of its own. Human double-grading with adjudication reduces ε for proof artifacts [4]. Each is a genuine advance on a specific term. None of them is an identification argument for Θ_{math}: none exhibits an intervention, a design, or an auxiliary measurement that separates a change in Θ_{math} from a change in η_M or in the instrument arguments of $\Phi_{\mathscr{B}}$. Identification of a latent trait from performance data requires an explicit measurement model together with evidence of invariance across conditions that ought not to matter—exactly the standard Messick sets for construct validity [7].

Three descriptions of $S_{\mathscr{B}}$ are worth holding simultaneously. In *statistical* language $\widehat{S}_{\mathscr{B}}$ is an estimator, with sampling variance, judge noise, and selection in $D_{\mathscr{B}}$; $S_{\mathscr{B}}$ is its estimand, the population functional it targets. In *mathematical* language it is a functional on model response distributions that factors through an artifact space. In *physical* language it is an observable—and the decisive question about any observable is whether it is invariant under the symmetry group of the target phenomenon. For mathematics that group is generated by the equivalences of Section 6; for $S_{\mathscr{B}}$, invariance has never been established. The three languages name one gap.

3.3 Reliability is not validity

A benchmark can be highly reliable and still not measure what its name says. Reliability—stability of the score under re-sampling, re-running, or re-grading—constrains ε in (5). Construct validity concerns whether $\Phi_{\mathscr{B}}$ responds to Θ_{math} rather than to η_M or to the instrument [7]. The two come apart in a specific and testable way: an instrument whose acceptance depends on the presentation of a problem can be perfectly reproducible on the canonical presentation and still fail to track any presentation-independent attribute. That failure mode is the one this paper makes measurable.

4 Cross-Cutting Construct-Validity Limits

This section states the limits that apply to *every* instrument in the corpus, independent of its individual design quality. The results are elementary by intent: their role is to convert informal complaints into statements with hypotheses and proofs, so that each can be met with evidence rather than rhetoric.

4.1 Artifact projection and collapse

Each benchmark converts a mathematical act into an artifact. Let $\mathscr{R}$ denote the space of reasoning processes—read modestly as complete traces, search trees, and intermediate derivations actually produced—and $\mathscr{A}_{\mathscr{B}}$ the observable artifact space. The conversion is a projection

$$\pi_{\mathscr{B}} : \mathscr{R} \longrightarrow \mathscr{A}_{\mathscr{B}}, \qquad J_{\mathscr{B}}(x,r) = j_{\mathscr{B}}\big(x, \pi_{\mathscr{B}}(r)\big). \tag{7}$$

The evaluator sees only $\pi_{\mathscr{B}}(r)$. Even when grading is perfectly accurate over artifacts it need not be faithful over reasoning: $\pi_{\mathscr{B}}$ is not injective, and its fibers are not observable through the artifact channel.

Definition 4.1 (Artifact collapse)**.** There may exist distinct $r_1, r_2 \in \mathscr{R}$ with $\pi_{\mathscr{B}}(r_1) = \pi_{\mathscr{B}}(r_2)$ although r_1 is a structurally correct, generalizing derivation and r_2 a memorized, gauge-fixed, or presentation-specific route to the same artifact. The fiber $\pi_{\mathscr{B}}^{-1}(a)$ mixes epistemically inequivalent processes, and any artifact-level score is constant on fibers by construction.

The corpus sits on a gradient of exposure—a decrease of fiber size that never reaches triviality. Final-answer grading collapses most strongly, which is exactly why MathArena introduced freshness [1] and MathConstruct complex witnesses [3]; verifier-checked constructions shrink the fiber but still identify all derivations of the same valid witness; proof text exposes far more structure but is then compressed into one bit [4]; kernel-checked terms verify derivability against a formal statement [2] without certifying that the statement captures the intended content (Section 4.6). At no point in the audited corpus does the fiber become a point.

Proposition 4.2 (Artifact non-identifiability)**.** *Let $\pi_{\mathscr{B}} : \mathscr{R} \to \mathscr{A}_{\mathscr{B}}$ be the artifact projection and suppose every score has the form $S(M) = \mathbb{E}[\, j_{\mathscr{B}}(\pi_{\mathscr{B}}(r))\,]$ for some judge $j_{\mathscr{B}}$ on artifacts. If there exist traces r_1, r_2 with $\pi_{\mathscr{B}}(r_1) = \pi_{\mathscr{B}}(r_2)$ but $I(r_1) \neq I(r_2)$ for a process-level functional $I : \mathscr{R} \to \mathbb{R}$, then no score depending on r only through $\pi_{\mathscr{B}}(r)$ distinguishes $I(r_1)$ from $I(r_2)$.*

Proof. Any such score is a function $g \circ \pi_{\mathscr{B}}$; since $\pi_{\mathscr{B}}(r_1) = \pi_{\mathscr{B}}(r_2)$ we get $g(\pi_{\mathscr{B}}(r_1)) = g(\pi_{\mathscr{B}}(r_2))$, while $I(r_1) \neq I(r_2)$. □

The proposition is trivial to prove and consequential to state: it converts "benchmark scores collapse reasoning" into an identifiability statement whose empirical content—whether frontier models actually populate the fibers with behaviourally distinct processes—is a question for measurement, not for argument. It also bounds this paper: IMR is itself a benchmark, its instrument also acts through a projection, and Proposition 4.2 applies to it verbatim (Section 7).

4.2 Finite-sample non-identification

Proposition 4.3 (Finite-sample non-identification)**.** *Let Θ be a parameter set, $\Theta_0 \subsetneq \Theta$ the finite tested subset with $\Theta \setminus \Theta_0 \neq \emptyset$, and $f : \Theta \to A$ a policy into an answer set A with $|A| \geq 2$. For any $\theta^* \in \Theta \setminus \Theta_0$ there exists a policy g with $g|_{\Theta_0} = f|_{\Theta_0}$ and $g(\theta^*) \neq f(\theta^*)$. Hence no finite test uniquely identifies behaviour outside the tested set.*

Proof. Define $g = f$ on $\Theta \setminus \{\theta^*\}$ and let $g(\theta^*)$ be any element of $A \setminus \{f(\theta^*)\}$, nonempty because $|A| \geq 2$. □

This does not show that models fail to generalize. It shows that generalization from finitely many tested variations—for instance from MathConstruct's sampled parameter set to possession of a uniform construction method [3]—is an inference requiring evidence or assumptions, and never follows from the finite record alone. The distinction is between *solving several instances* and *possessing a uniform procedure*, and it is the reason the design of Section 9 reports orbit-level rather than instance-level statistics.

4.3 Verifier-selection of the task distribution

Definition 4.4 (Verifier-selection bias)**.** Fix a declared evaluator class $\mathscr{E}$ and budgets: cost C, tolerated error ε, latency T, reliability α. Let

$$\mathscr{P}_{\text{eval}}(\mathscr{E};C,\varepsilon,T,\alpha) = \left\{ p \in \mathscr{P} : \exists E \in \mathscr{E} \text{ with cost} \le C,\ \text{error} \le \varepsilon,\ \text{latency} \le T,\ \text{reliability} \ge \alpha \right\}. \quad (8)$$

Verifier-selection bias is the distortion induced when $D_{\mathscr{B}}$ is supported mainly on $\mathscr{P}_{\text{eval}}$ rather than on a declared target domain $\mathscr{P}_{\text{target}}$.

$\mathscr{P}_{\text{eval}}$ is not absolute: it is relative to $\mathscr{E}$, to the budgets, and to the declared target, and it expands as evaluation methods improve. The construct is therefore a design-dependent selection mechanism, and this paper makes it auditable rather than rhetorical: for a target domain declared in advance, sample a task list from domain experts and report the fraction admitting a feasible evaluator under stated budgets, together with the excluded strata. We call this the *coverage audit*, and we recommend it as a reporting item (Section 10).

Easily scored: exact numerical answers, finite combinatorial objects, matrices under explicit constraints, short competition proofs, statements already formalized in a library, false claims with clean counterexamples. Hard to score reliably at scale: arguments turning on tacit domain conventions, global geometric obstructions, conceptual reformulation, definition choice, conjecture generation, regulator-dependent physical reasoning, equivalence across dual formulations, explanatory depth. That the second list is under-represented is not a defect of any team's judgment—a benchmark cannot evaluate what it cannot observe—but it does mean that a term such as "mathematical reasoning" is being operationalized on the subset selected by gradability, and the substitution must be made explicit rather than absorbed into the name. The corpus supplies internal evidence for the strictness of the inclusion: ProofRank reports abandoning proof-based items whose scalable validation was infeasible [6], and the platform records that its research tier measures a proxy for the intended construct [2].

Physical intuition. This is the evaluation-theoretic analogue of an experimental acceptance cut. A detector with acceptance $\varepsilon(p)$ measures not $\mathrm{d}\sigma$ but the reweighted, renormalized $\varepsilon\,\mathrm{d}\sigma / \int \varepsilon\,\mathrm{d}\sigma$; conclusions drawn without unfolding the acceptance are statements about the apparatus as much as about nature. Here $\varepsilon = \mathbf{1}_{\mathscr{P}_{\text{eval}}}$ is a hard cut, and none of the audited instruments reports such an unfolding or a coverage correction.

4.4 Judge substitution

Where human grading does not scale, an LLM judge is substituted, and the substitution is validated by agreement with human labels on a sample. Agreement is a necessary check and an insufficient one, for two reasons. First, agreement identifies accuracy only under an annotation model: with correlated errors between judge and humans—shared surface preferences, shared misreadings of a convention—high agreement is compatible with correlated inaccuracy, which is why latent-annotator models rather than majority vote are the appropriate aggregation [31]. Second, agreement is measured on the canonical presentation. A judge that accepts a correct proof in one notation and rejects the same content in another has a defect that no agreement statistic on a fixed presentation can reveal. This motivates treating the

evaluator as part of the apparatus and auditing it before any model-level claim (Section 9); it is also the one place where this paper's own instrument is most exposed, and it is gated accordingly.

4.5 Aggregation and dimensionality

Two different objects are easily confused here: a *composite index*, which is a declared weighted summary of separate measurements, and a *reflective latent-variable model*, in which one underlying trait is posited to generate the observed responses. The first needs transparent weights and an interpretation; the second additionally needs its measurement assumptions examined. Aggregation as such does not imply the second, and the criticism that follows applies only where the second is asserted.

The MathArena platform is the instrument in the corpus that takes this step explicitly, and its choice deserves to be described accurately. To summarize performance when not every model is run on every benchmark, the platform fits a two-parameter item-response model in which each model m receives a scalar capability parameter $\theta_m \in \mathbb{R}$, each question q a difficulty d_q and a discrimination α_q, with

$$p_{mq} = \sigma\big(\alpha_q(\theta_m - d_q)\big), \tag{9}$$

estimated by maximum likelihood; expected performance is the average of p_{mq} over questions, and the same fit is used to impute unobserved model–benchmark cells. The platform also reports robustness checks: a one-parameter variant with common discrimination yields a nearly identical ranking (Spearman 0.99), and recomputation after removing one benchmark family at a time leaves nine of the top ten models in the top ten, with a largest rank shift of five positions [2].

This is a real measurement model, and the critique must therefore be made at the right level. What the reported checks establish is *rank stability of a summary statistic* under a nearby link specification and under family ablation. What they do not establish is what a latent-trait interpretation of θ_m requires: substantive unidimensionality; local independence of items given θ_m; measurement invariance across the heterogeneous tiers, so that a final-answer item, a graded proof, a corrupted-claim response, and a kernel-checked term are commensurable indicators of one trait; absence of differential item functioning across model families with different training and tool access; and the missing-at-random assumption implicit in imputing unobserved cells from the same fit. A Spearman correlation of 0.99 between a 2PL and a 1PL ranking is evidence that the ranking is insensitive to the discrimination parameters, not evidence that one latent dimension generated the responses; and leave-one-family-out stability is evidence that no single family drives the ranking, not evidence that the families measure the same thing. Under a genuinely multidimensional response process a unidimensional fit still returns a well-defined $\hat{\theta}_m$, and its ranking can remain stable under both perturbations; that possibility is what the reported checks cannot rule out.

Mathematical competence is plausibly multidimensional—final-answer accuracy, constructive witness production, proof correctness, formalization, resistance to false premises, proof quality, and representation stability need not co-vary—and the platform itself notes that no single benchmark spans the full range of mathematical capabilities [2]. The gap is therefore precise: a fitted unidimensional model exists, and the evidence that would license reading its scalar parameter as *the* mathematical capability—dimensionality analysis, invariance testing, DIF diagnostics, and out-of-sample predictive validation—is not reported by the platform, and is not supplied by this paper either.

4.6 Formalization adequacy

Proposition 4.5 (Bounded formalization adequacy)**.** *Let $\mathscr{D}$ be a sound proof checker, φ a formal statement, and π a candidate term. Then* $\mathrm{Check}_{\mathscr{D}}(\pi,\varphi) = 1$ *certifies derivability of φ in $\mathscr{D}$, i.e. inhabitation of the corresponding type. Adequacy of φ to an informal target $\varphi_{\text{intended}}$ is not thereby certified: it is a relation between a formal object and an object outside $\mathscr{D}$'s object language, and its evaluation requires an externally supplied interpretation $\mathscr{I}$. If instead a* formal *specification σ is supplied, then an extended*

system $\mathscr{D}'$ can certify a refinement or equivalence $\varphi \simeq \sigma$; this relocates the adequacy obligation to σ, and does not discharge it.

The proposition is deliberately weaker than an impossibility claim, and the weaker statement is the true one: kernel acceptance does not *by itself* establish the interpretation contract that fixes what the formal statement is supposed to mean. A broader system may well propose candidate formalizations, refinement maps, or specifications; generating such a contract and validating it against the informal source are separate obligations, and neither is discharged by the kernel's verdict. Under propositions-as-types [12], formalizing means choosing a type, and a verified proof certifies an inhabitant of that type; if the chosen type encodes a gauge-fixed representative where the mathematics concerns the quotient object, the verification is sound and the construct validity is void—the right theorem about the wrong type. Two canonical illustrations: a kernel-checked statement in which the intended convention for the natural numbers is mistranslated, so that what is verified is not the intended theorem [11]; and a gauge-fixed potential formalized where the moduli object $[A] \in \mathscr{A}/\mathscr{G}$ was meant (Section 8). Formalization transfers part of the trust problem from proof checking to specification checking, which is precisely the architecture small-kernel systems are designed around [15].

5 Instrument-by-Instrument Audit

Each subsection answers the same six questions: (Q1) what capability language is used; (Q2) what is directly observed; (Q3) what evaluator and protocol produce the label; (Q4) what inference the design licenses; (Q5) which broader inference is not identified; (Q6) what additional evidence would close the gap. Capability language was quoted in Section 2.1 and is not re-quoted here. Q4 is stated before Q5 throughout, and the concession in Q4 is meant seriously: in every case the instrument establishes something that was not previously established.

5.1 MathArena

Q1–Q2. MathArena describes itself as a rigorous and up-to-date evaluation of mathematical reasoning [1]. What it records is correctness on competition problems released after the evaluated models' cutoffs, together with graded proof-writing performance on proof-based competitions.

Q3. Final answers are handled by automated rule-based extraction and comparison—a symbolic parser with a re-ask safeguard when the output format is malformed—and proofs are scored by human graders against a rubric. The protocol fixes prompting, sampling, and attempt counts across models—an important design property, since (3) makes the protocol part of what is measured.

Q4. The design licenses a useful claim under an explicit and plausible identifying assumption: *if* the competition was released after the model's training cutoff and the released material did not enter the model's context by other routes, then performance on it cannot be explained by verbatim exposure to those problems. The instrument also reports markedly weaker performance on proof-based competitions than on final-answer ones—evidence that cuts against naive capability readings of final-answer scores, and is to the instrument's credit. The sharper, matched comparison—the same problems scored for answer and for proof—is what the Open Proof Corpus supplies [4]; the cross-competition contrast alone confounds the answer/proof distinction with differences in problem difficulty.

Q5. Freshness is a *temporal* property. It certifies that the specific problem is new to the model; it does not certify that the model's correctness is stable under reformulation of the same content, nor does it exclude the non-memorization components of η_M in (5)—template transfer from structurally similar training problems, search-and-check strategies, or protocol adaptation. Freshness and semantic stability are logically independent properties: an instrument can have either without the other. Concretely, a model that solves a fresh problem in its canonical coordinates and fails the same problem in a certified

change of coordinates has passed the freshness test and failed the invariance test, and the published score cannot distinguish it from a model that passes both.

Q6. Add paired presentations of the *same* fresh problems under certified equivalences, judge the transformed arm against a transported reference answer, and audit the grading pipeline for its own presentation sensitivity before attributing any difference to models. Report the invariance axis separately rather than folding it into a single average.

5.2 The MathArena platform

Q1–Q2. The platform is presented as a resource for accurate, up-to-date evaluation of mathematical capability [2], and aggregates heterogeneous tiers: competition problems, research-derived problems, corrupted research claims, and formal statements. It additionally reports an *expected performance* summary derived from the fitted item-response model (9).

Q3. Each tier carries its own evaluator—automated rule-based answer extraction and comparison for final-answer tiers, human or model grading for proofs, corrupted-claim classification, kernel acceptance for formal statements—and the platform reports them together, with unobserved model–benchmark cells imputed from the same item-response fit.

Q4. The breadth is a genuine advance: it is the only instrument in the corpus that spans final answers, proofs, false-claim resistance, and formalization within one reporting frame. It is also the only one that makes its aggregation model explicit rather than implicit—a scalar capability parameter, difficulty and discrimination per question, maximum-likelihood estimation, and published robustness checks against a one-parameter variant and against family ablation. And it states its own coverage limits, including that problem-solving performance omits interaction, conjecturing, and communication. Making the measurement model explicit is the right methodological move, and it is what allows the model to be audited at all.

Q5. What is not identified is the reading of the fitted scalar as a broad mathematical capability. The published robustness evidence concerns rank stability under a nearby link specification and under family ablation; the assumptions a latent-trait interpretation needs—substantive unidimensionality, local independence, measurement invariance across tiers as different as final answers and kernel-checked terms, absence of differential item functioning across model families, and missingness-at-random for the imputed cells—are not tested, and a unidimensional fit to multidimensional data can still produce a well-defined and stable ranking (Section 4.5). Separately, the research and formal tiers inherit the adequacy gap of Proposition 4.5: kernel acceptance certifies derivability relative to a formalized statement in a fixed library, and the step from there to the informal research theorem is an interpretation, not a verification.

Q6. Report the diagnostics that the latent-trait reading requires: dimensionality analysis on the item-by-model response matrix, local-independence residual checks, cross-tier invariance and DIF tests by model family, sensitivity of expected performance to the missingness mechanism, and held-out predictive validation on a tier withheld from the fit. Add a representation-stability axis as a separate coordinate rather than as another indicator of the same θ; and for the formal tier publish the source-to-formal interpretation contract alongside each formalized statement, so that the adequacy obligation is visible and auditable rather than implicit.

5.3 MathConstruct

Q1–Q2. MathConstruct is framed as an evaluation of LLM reasoning capabilities through constructive proofs [3]. What is recorded is whether a submitted object is accepted by a problem-specific executable verifier, together with success across generated parameter variations.

Q3. The verifier is a program; the protocol includes parameter regeneration and a robustness criterion requiring success across several variants.

Q4. This is the strongest correctness channel in the corpus: verifier acceptance of a nontrivial witness is hard to obtain by guessing, free of grader variance conditional on the verifier's correctness and specification adequacy, and the parameter-variation design directly addresses instance-level memorization. Robust accuracy over variants is a real advance over single-instance scoring.

Q5. Two inferences remain open. First, by Proposition 4.3, success on sampled parameters does not identify possession of a uniform construction method: policies agreeing on the tested set can disagree everywhere else, so the robustness criterion bounds instance-level brittleness without establishing uniformity. Second, and more fundamentally for capability language, verifier acceptance is silent about the response process: a witness found by enumeration, by pattern completion from a structurally similar solved problem, or by first-principles construction is the same artifact, and by Proposition 4.2 no artifact-level score separates them. The benchmark measures witness production, which is a real ability; the inference to reasoning-as-derivation is the unlicensed step.

Q6. Where a uniform method is claimed, require an artifact that exhibits it—a general construction with its correctness argument, or a program parameterized by the family—and grade the argument, not only the instance. For the response-process question, pair the construction task with certified reformulations that preserve the object but change the presentation, and test whether acceptance survives.

5.4 The Open Proof Corpus

Q1–Q2. The OPC frames contamination control as a challenge for evaluating models' true reasoning capabilities [4], and records human correctness judgments—with LLM-judge comparisons—on thousands of natural-language proofs.

Q3. Human graders working from a rubric, with multiple judgments per proof, plus an evaluation of LLM judges against those labels.

Q4. The corpus establishes something important and previously only suspected: a correct final answer does not imply a correct proof, and the conditional probability of a valid proof given a correct answer varies substantially across models. It also quantifies the informal–formal gap. These are among the most decision-relevant facts in the whole corpus, and they are established by the design.

Q5. Three inferences are not identified. First, judge accuracy: agreement between an LLM judge and human graders identifies accuracy only under an annotation model with uncorrelated errors, and the required latent-annotator analysis is not what agreement statistics report (Section 4.4). Second, the binary correctness label compresses a structured obligation set: two proofs both labelled incorrect may fail for reasons as different as a missing convergence hypothesis, an ignored boundary term, and a claimed global conclusion from a local lemma, and the single bit records none of this. Third, the distribution is competition mathematics; the corpus itself notes that research-level mathematics lies outside it, so the extension from graded competition proofs to proof competence in working mathematics is not supported by the design.

Q6. Replace the scalar label with a structured obligation record (Section 9) so that failure modes are separable; fit a latent-annotator model rather than treating agreement as truth; and audit judges on presentation-equivalent rewritings of accepted proofs, which is the specific failure this design cannot currently see.

5.5 BrokenMath

Q1–Q2. BrokenMath scopes itself narrowly and accurately, to sycophancy in theorem proving [5]; it records whether a model endorses, corrects, detects, or resists expert-refined false statements.

Q3. A corruption pipeline generates plausible falsifications of true theorems with expert refinement; responses are classified into behavioural categories.

Q4. The design licenses exactly what it claims: a rate of sycophantic endorsement under that corruption distribution, together with the demonstration that strong models produce convincing proofs of false statements. Because the claim is scoped to a behaviour rather than to a capability, the interpretation gap that dominates the other subsections is largely absent here. This is the corpus's cleanest match between claim and observable.

Q5. The residual limit is coverage, not overclaim. The corruption process defines the space of falsehoods the instrument can see. Falsehoods that are locally impeccable—a solution that exists locally but does not glue, a gauge-variant quantity read as an observable, an interchange of limit and integral without a convergence hypothesis, a discarded boundary term, a classical symmetry assumed to survive quantization—are not surface corruptions of a statement and are largely outside the generated distribution. Resistance measured on one falsehood distribution does not transfer to another, and the structurally hardest falsehoods are precisely the ones where every displayed line looks correct.

Q6. Extend the corruption family to obstruction-level falsehoods with certified anti-equivalence witnesses (Section 9), and separate the two mechanisms of false endorsement—failure to detect versus compliance under pressure—by contrasting prompt conditions that request a proof, ask whether the claim is true, request a counterexample, or express skepticism.

5.6 ProofRank

Q1–Q2. ProofRank introduces its five dimensions explicitly as scalable proxies of proof quality [6]: conciseness, computational ease, cognitive simplicity, diversity of method, and adaptivity to a requested technique. What is recorded is judge assessment on those dimensions.

Q3. Model and human judgments under dimension-specific rubrics.

Q4. The move beyond binary correctness is correct and overdue—not all correct proofs are equally useful—and the explicit labelling of the dimensions as proxies is the most careful construct statement in the corpus. This paper does not argue against a claim ProofRank does not make.

Q5. What remains open is whether the proxies themselves are stable under the transformations that ought not to matter. A proof judged concise in one notation may be judged verbose in an equivalent one; a proof judged cognitively simple may owe its simplicity to a coordinate choice that conceals the structural obligation; and quality judgments delivered by a judge are subject to the same presentation sensitivity as correctness judgments. In addition, there is a domain-specific structural-validity layer—whether an argument is manifestly covariant, gauge-invariant, functorial, and stable under the relevant duality—that is not represented among the five proxies. This is not offered as a general axis of proof quality; it is a layer that matters in mathematical physics and adjacent fields, and it is simply outside what ProofRank set out to measure.

Q6. Measure proxy stability directly: present the same proof content in certified-equivalent presentations and report the variance of each dimension across the orbit, together with judge-substitution sensitivity. Add the structural-quality dimensions as a separate, declared layer rather than as an implicit component of simplicity.

5.7 What the audit shows

Across the six instruments the pattern is stable and the diagnosis is uniform. Each design supports its Q4 claim. In every case where capability language is used, the step from Q4 to that language crosses at least one of the gaps of Section 4: artifact projection (MathConstruct, all final-answer tiers), finite-sample non-identification (parameter-variation robustness), verifier-selection of the distribution (all), judge sub-

stitution (OPC, ProofRank, platform tiers), a fitted latent-trait index whose interpretive assumptions are untested (platform), or formalization adequacy (formal tiers). None of these gaps is automatically closed merely by a better model, a larger sample, or a fresher competition. Each requires a different kind of evidence: an intervention, a measurement model, a coverage audit, an annotation model, or an invariance test.

The rest of this paper supplies one of them—the invariance test—in full technical detail, and states plainly that supplying one does not supply the others.

6 Certified Semantic Transport

Section 4 identified several gaps. This section develops the apparatus for one of them: whether accepted correctness descends from surface representatives to semantic equivalence classes. The construction is stated carefully, because the naive version—"the model should give the same answer to equivalent problems"—is false as stated: under a change of basis the coordinates of a vector must change, and an instrument demanding literal answer identity would score the correct response wrong.

6.1 Presentations, certified equivalences, and what is excluded

Let $\mathscr{X}$ be a space of problem presentations. Mathematics is not a free syntactic monoid of presentations; it is structured by equivalence. Write $x \sim x'$ when x and x' present the same content under an accepted transformation. Admissible generators include change of notation; isomorphism of underlying structures; gauge transformation $A \mapsto A^g$; established duality within its domain of validity [21, 25]; reparametrization and change of coordinates; and change of local trivialization of a bundle.

Definition 6.1 (The certified-equivalence groupoid)**.** $\mathscr{G}_{\mathrm{IMR}}$ has as objects the presentations $x \in \mathscr{X}$, and as morphisms $T : x \to x'$ the *certified semantic equivalences*: transformations accompanied by a certificate recording the transformation applied, the declared domain on which it is valid, the validation that it preserves the mathematical content in question, and the induced action on answers (Definition 6.2). Composition is composition of transformations, identities are trivial re-presentations, and—this is a restriction, not an observation—*every morphism admitted to $\mathscr{G}_{\mathrm{IMR}}$ is required to be invertible on its declared domain*, with the inverse certificate exhibited.

The invertibility requirement excludes three families of passage that earlier formulations of this framework wrongly admitted, and the exclusion matters for the confirmatory argument:

- *Formal–informal translation.* The passage from an informal statement to a formalization is not invertible: distinct informal readings can formalize to the same type, and Proposition 4.5 is exactly the statement that the reverse direction requires an external interpretation. This passage belongs to a separate *translation layer*, represented by spans or a category of partial maps, and no confirmatory quantity in this paper is computed over it.
- *Restricted or conjectural dualities.* A duality dictionary is admitted only where it is mathematically established, and transport is graded only within the dictionary's declared domain of validity [25]. Conjectural correspondences are excluded from $\mathscr{G}_{\mathrm{IMR}}$ entirely.
- *Foundation changes.* A classical theorem and its constructive counterpart are not automatically the same statement; a result provable in one foundation may require modification in another. Foundation contrasts are studied in a separate layer and never treated as certified equivalences.

Equivalences of categories are deliberately excluded from v1. Equivalence, not isomorphism, is the appropriate notion of sameness for structured mathematics [14]—and that is precisely why it cannot be

admitted here without changing the ambient structure. An equivalence is not an invertible arrow on the nose: it carries a quasi-inverse together with unit and counit natural isomorphisms, and the inverse laws hold only *up to* natural isomorphism. Choosing a specific adjoint-equivalence datum does not repair this, because the chosen datum is still not a strictly invertible 1-cell; and passing to the core of **Cat** retains strict categorical isomorphisms, not equivalences. An earlier version of this framework claimed the second manoeuvre; it does not work, and the claim is withdrawn.

IMR v1.0 therefore admits strict isomorphisms only. Where two presentations differ by an isomorphism of the structures involved—relabelled basis, renamed generators, an isomorphic presentation of a group—the isomorphism is an arrow of $\mathscr{G}_{\mathrm{IMR}}$ and carries a genuine bijection on answers. Where they differ by a non-strict equivalence, the pair is *not* admitted to the confirmatory groupoid. Handling those honestly requires a bicategory or $(2,1)$-category in which the natural isomorphisms are certified 2-cells, and the resulting transport is defined only up to coherent 2-cell; that is a v2 object, and nothing in the design of Section 9 depends on it. The governing intuition—that equivalent structures should be interchangeable in reasoning—is the informal content of the univalence principle [13], adopted here as a design criterion imposed by fiat, not as a theorem transported from homotopy type theory, and not as a licence to treat equivalences as isomorphisms.

Two notational conventions are fixed for the whole paper: a generic group is never written $\mathscr{G}$ (that symbol is reserved for the groupoid), and T is never a sampling temperature (it is always a transformation; temperature, where needed, is ϑ inside Γ).

6.2 The answer functor and the correctness subfunctor

Definition 6.2 (Answer functor, correctness subfunctor, transport)**.** Answer spaces come in two kinds, and only one of them supports a bijective transport. A *final-answer* space carries typed values—integers, rationals, tuples, matrices, spectra, labels—for which a certified transformation determines a bijection. A *proof-artifact* space carries derivations, for which no canonical bijection between presentations exists and none is claimed; proofs are handled by Definition 6.5 below. The functor construction is stated for final-answer spaces.

An *answer assignment* for $\mathscr{G}_{\mathrm{IMR}}$ is a functor into the groupoid of sets and bijections,

$$A:\ \mathscr{G}_{\mathrm{IMR}} \longrightarrow \mathbf{Set}_{\cong}, \qquad x \longmapsto A_x, \qquad (T : x \to x') \longmapsto \tau_T := A(T) : A_x \xrightarrow{\sim} A_{x'}, \tag{10}$$

so that functoriality reads

$$\tau_{U\circ T} = \tau_U \circ \tau_T, \qquad \tau_{T^{-1}} = \tau_T^{-1}, \qquad \tau_{\mathrm{id}_x} = \mathrm{id}_{A_x}. \tag{11}$$

A correctness assignment cannot be a subfunctor of A itself, because $\mathbf{Set}_{\cong}$ has only bijections as morphisms and the inclusions $C_x \hookrightarrow A_x$ are not bijections. Write therefore

$$U : \mathbf{Set}_{\cong} \hookrightarrow \mathbf{Set} \tag{12}$$

for the inclusion that forgets invertibility, so that $U \circ A : \mathscr{G}_{\mathrm{IMR}} \to \mathbf{Set}$ is a **Set**-valued functor with the same objects and the same action on arrows. A *correctness assignment* is a subfunctor

$$C \hookrightarrow U \circ A \quad \text{in } [\mathscr{G}_{\mathrm{IMR}}, \mathbf{Set}], \tag{13}$$

that is, a presentation-indexed family of nonempty sets $C_x \subseteq A_x$ for which the naturality square of the inclusion commutes—equivalently, for which every certified transformation preserves correctness,

$$\boxed{\tau_T\big(C_x\big) = C_{x'} \quad \text{for every certified } T : x \to x'.} \tag{14}$$

When the acceptable answer is unique, $C_x = \{a_x\}$, condition (14) reduces to an *equivariant section a* of A:

$$a_{x'} = \tau_T(a_x), \tag{15}$$

and $\tau_T(a_x)$ is the *transported reference answer* against which the transformed arm is judged. A transformation is admitted to $\mathscr{G}_{\mathrm{IMR}}$ only if its certificate exhibits τ_T and verifies (14); this is what "certified" means. Exhibiting τ_T means supplying a *total bijection* on a declared source domain together with its inverse, a transport identifier and version, the source and target answer types, and a certificate digest—not merely naming the image of the reference answer. Three conditions are verified at admission: $\tau_T^{-1}(\tau_T(x)) = x$ on the certified source, $\tau_T(\tau_T^{-1}(y)) = y$ on the certified target, and—the condition the first two do not imply—$\tau_T(A_x) = A_{Tx}$, image equal to declared target. A declared target of $\{2,6\}$ for $v \mapsto 2v$ on $\{1,2\}$ satisfies both round-trip laws pointwise while the image is $\{2,4\}$. A further distinction is made explicit, because it is the difference between a bijection and a probe: a transport declares either a *finite* answer space, in which the enumerated set *is* the space, or a *typed* space described by predicates, in which the enumerated values are only probes and τ_T is total on the type. A well-typed but unexpected answer must transport and produce a defect, never raise.

For a typed space, extensional checking on a finite set is not a certificate. A branch firing at one large value is invisible to symbolic evaluation and outside any bounded sample, yet breaks the inverse law there. The resolution is architectural, not statistical: production typed transports are *package-owned symbolic specifications* from a closed set, their executable maps are *generated internally* from those expressions, and $\tau^{-1}(\tau(x)) = x$ and $\tau(\tau^{-1}(x)) = x$ are verified *symbolically* on the declared type. Confirmatory mode accepts no caller-supplied transport function, so faithfulness of an external implementation cannot arise. Each issued transport carries an opaque capability token held only by the sealed registry; admission checks that token, the executable maps, the declared types and the frozen digest—never a flag a caller can set. Two digests stay separate: the *transport* certificate (registry-level, immutable) attests a total bijection with the declared inverse; the *arrow* certificate attests that the transformation is a valid equivalence and this transport its induced answer action. When the source and target answer *types* coincide the map must be an automorphism of that type, which is why $v \mapsto 2v$ is inadmissible as $\mathbb{Z} \to \mathbb{Z}$ and admissible only when the target type is declared as $2\mathbb{Z}$. A certificate that fixes only $\tau_T(a_x)$ determines the transported gold and nothing else, so it cannot decide whether a model's non-gold answer commuted; the released implementation therefore refuses to compute a discrepancy from a partial map, and every arrow record carries the total map it was graded under.

This formulation replaces an earlier one in which ground truth was a functor into a codiscrete answer category. That construction does not do the work required of it: a codiscrete target has exactly one morphism between any two objects, so every assignment is functorial and no constancy on orbits is forced, while the opposite choice—a discrete target—would forbid the non-identity transport that the change-of-basis example makes mandatory. The functor into $\mathbf{Set}_{\cong}$ with an equivariant subfunctor is the correct object: it carries genuine bijections on answers and states correctness as a condition those bijections must respect.

Remark 6.3 (Functoriality does not give path independence)**.** Equations (11) constrain composites; they do *not* force distinct parallel morphisms to agree. If $T, T' : x \to x'$ are two certified arrows between the same presentations, in general $\tau_T \neq \tau_{T'}$, since $\tau_{T'}^{-1} \circ \tau_T = \tau_{T'^{-1} \circ T}$ is the action of a loop at x, and the isotropy group $\mathrm{Aut}_{\mathscr{G}_{\mathrm{IMR}}}(x)$ may act non-trivially on A_x. Two consequences are binding for measurement. First, *every reported quantity is indexed by an arrow*, never by an unordered pair of presentations: the transported reference answer is $\tau_T(a_x)$ for the specific certified T shipped with the item. Second, the correctness condition (14) applied to loops is exactly the requirement that the isotropy action preserve the correctness set,

$$\tau_\gamma(C_x) = C_x \qquad \text{for all } \gamma \in \mathrm{Aut}_{\mathscr{G}_{\mathrm{IMR}}}(x), \tag{16}$$

a nontrivial, checkable coherence condition on the item bank rather than a consequence of functoriality. Where several certified routes between two presentations are shipped, (16) must be verified for each generated loop; where τ is the identity on all admitted arrows the condition holds trivially, and it becomes binding exactly when non-identity transports enter (Section 9.8). Path independence, if wanted as a global property, must be imposed and certified—or specified at the level of certified 2-cells in a bicategorical refinement—never inferred.

6.3 Normalization, quotient validity, and the equivariance condition

Definition 6.4 (Semantic normalizer)**.** For each presentation fix

$$\Pi_x : \{\text{raw model transcripts on } x\} \longrightarrow A_x \sqcup \{\mathsf{parse\text{-}fail}\}, \tag{17}$$

which extracts either a well-typed answer or an explicit typed parse failure—never a silent coercion of unparseable output into a wrong answer. Π_x normalizes the *final typed answer together with its declared coordinates, conventions, and units*; everything else—whether a proof discharges its structural obligations—is left to the judge and recorded separately (Section 9). Transport is never applied outside A_x, and it is not extended to the parse-failure symbol: an extension sending $\mathsf{parse\text{-}fail} \mapsto \mathsf{parse\text{-}fail}$ would make two failed parses satisfy the equivariance equation exactly, so that "equivariant" could describe a pair of non-answers. Semantic equivariance is therefore defined only on the *joint-parse event*

$$\mathrm{JP}(T) = \big\{\Pi_x(M(x)) \in A_x\big\} \cap \big\{\Pi_{Tx}(M(Tx)) \in A_{Tx}\big\}, \tag{18}$$

and refusal or parse failure on either arm is handled exclusively as a response-process defect in the composite estimand of Section 9.

Definition 6.5 (What is measured on a proof artifact)**.** Proof artifacts are *not* compared as objects. A relation between spaces of derivations strong enough to support a correctness claim would need acceptable-proof subsets, an identity axiom, exact two-sided correctness preservation, and a normalizer from raw transcripts into a space of proof objects; specifying that structure—plausibly a **Rel**-valued lax functor—is deferred to v2, and an earlier version of this framework asserted such a relation without supplying it. In v1 the measured object on a proof task is the pair

$$\mathrm{summ}_x(y) \;=\; \big(\ \mathrm{claim}_x(y) \ \in A_x\ ,\ \ \mathbf{J}(x,y) \in \{0,1,\mathsf{NA},?\}^9\ \big), \tag{19}$$

the *transported final claim* together with the *obligation vector* of Definition 9.3. The claim lives in a typed answer space and transports bijectively under τ_T. The obligation vector is transported by an explicit map supplied in the certificate. Write $\mathscr{I} = \{1,\dots,9\}$ for the coordinate index set and $m_x \subseteq \mathscr{I}$ for the applicability mask of presentation x. Since $\mathscr{G}_{\mathrm{IMR}}$ consists of strictly invertible arrows, the obligation action must be strictly invertible too, and that forces a decomposition of the mask before any map is written down. Partition

$$m_x \;=\; m_x^{\mathrm{inv}} \;\sqcup\; m_x^{\mathrm{pres}}, \tag{20}$$

where m_x^{inv} holds the *equivalence-invariant* obligations—those the certificate proves are present on both sides of every arrow in the declared class—and m_x^{pres} holds the *presentation-specific* ones, which exist only relative to a particular presentation. The certificate of T then supplies a *bijection on the invariant part alone*,

$$\lambda_T : m_x^{\mathrm{inv}} \xrightarrow{\ \sim\ } m_{Tx}^{\mathrm{inv}}, \tag{21}$$

and the three laws are then exactly the functor laws, holding on the nose:

$$\lambda_{\mathrm{id}_x} = \mathrm{id}, \qquad \lambda_{U\circ T} = \lambda_U \circ \lambda_T, \qquad \lambda_{T^{-1}} = \lambda_T^{-1}. \tag{22}$$

Presentation-specific obligations are *not* acted on by the groupoid. They are recorded as source and target annotations, reported per arm, and excluded from the transported comparison. An earlier version of this specification allowed λ_T to retire coordinates to a symbol † and to leave target coordinates without preimages; that is incompatible with (22) under strict invertibility—a retired coordinate cannot be recovered by $\lambda_{T^{-1}}$, several retired coordinates cannot map injectively to one symbol, and an introduced coordinate has no source—so retirement and introduction are exactly what (20) moves out of the group action. A framework that genuinely needs them must be lax or non-invertible, and that is a v2 object. The proof-summary discrepancy is defined on the invariant coordinates only. Let

$$K_T = \big\{\, i \in m_x^{\mathrm{inv}} \;:\; J_i(x) \in \{0,1\} \text{ and } J_{\lambda_T(i)}(Tx) \in \{0,1\} \,\big\} \tag{23}$$

be the set of *comparable* coordinates: those applicable and resolved on both sides. Coordinates carrying ? on either side are routed to adjudication and, if unresolved, excluded from K_T and reported as adjudication load; NA coordinates never enter. Then the *claim* and *obligation* components are computed separately,

$$\delta_T^{\mathrm{cl}} = \mathbf{1}\big\{\mathrm{claim}_{Tx} \neq \tau_T(\mathrm{claim}_x)\big\}, \qquad \delta_T^{\mathrm{ob}} = \begin{cases} \mathsf{unavailable}, & K_T = \emptyset, \\ \mathbf{1}\big\{\exists\, i \in K_T :\; J_i(x) \neq J_{\lambda_T(i)}(Tx)\big\}, & \text{otherwise}, \end{cases} \tag{24}$$

and combined so that *a claim mismatch is decisive on its own*:

$$\delta_T^P(M) = \begin{cases} 1, & \delta_T^{\mathrm{cl}} = 1, \\ \mathsf{unavailable}, & \delta_T^{\mathrm{cl}} = 0 \text{ and } K_T = \emptyset, \\ \delta_T^{\mathrm{ob}}, & \text{otherwise}. \end{cases} \tag{25}$$

The ordering matters and an earlier version had it backwards: branching first on $K_T = \emptyset$ returned unavailable even when the transported claim definitely mismatched, which contradicts Definition 7.1, where claim equality is an independent requirement. Only the *obligation* component can be unavailable, and unavailable is a declared third outcome rather than a silent zero: with no comparable coordinate that channel carries no information, the claim comparison stands alone, and the arm is counted in the adjudication ledger. Presentation-specific coordinates are reported per arm and never enter δ_T^P. This is what "the certificate relabels the obligations" means, and without (21)–(22) the phrase would be an intention rather than a definition. Consequently equivalent presentations may elicit entirely different proof texts, strategies, and lemma inventories at no penalty: what is compared is the claim and the discharged obligations, never a derivation as an object or a string.

Definition 6.6 (Quotient validity)**.** For each presentation let $V_x : A_x \to \{0,1\}$ be the verdict map $V_x(\alpha) = \mathbf{1}\{\alpha \in C_x\}$, defined on well-typed answers only; parse failure is not a verdict and is handled in the response-process layer by the missingness rule of Section 9. The evaluation is *quotient-valid* on the orbit of x when

$$V_{x'} \circ \tau_T \;=\; V_x \qquad \text{for every certified } T : x \to x', \tag{26}$$

i.e. when V is a natural transformation

$$V :\; U \circ A \;\Longrightarrow\; \Delta_{\{0,1\}} \tag{27}$$

in $[\mathscr{G}_{\mathrm{IMR}}, \mathbf{Set}]$, from the **Set**-valued functor $U \circ A$ to the constant functor at $\{0,1\}$. The forgetful U is what makes this well typed: the components V_x are not bijections, so they are not morphisms of $\mathbf{Set}_{\simeq}$, and an earlier version of this framework nevertheless called them a natural transformation out of A. Condition (26) is equivalent to (14) and, unlike the informal demand that "the judge give the same verdict on equivalent problems," it is type-correct: both sides are maps out of A_x.

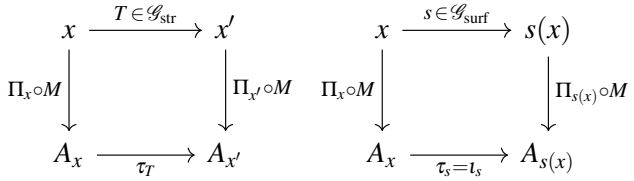


Figure 1: The measured object. Left: a certified *structural* arrow, whose square commutes exactly when the model is transport-equivariant; the defect δ_T is the failure of commutation. Right: the matched certified *surface* control, an arrow of the same groupoid whose certificate asserts presentation-level re-expression and declares the canonical identification ι_s of answer spaces (the identity whenever the two spaces coincide by construction). The confirmatory endpoint is the difference of the two defects, so that ordinary rewording brittleness is subtracted rather than counted. Before either square is attributed to the model, the same comparison is run with M replaced by expert reference answers to audit the evaluator (Definition 9.5).

The requirement applies to judged semantic content, not to raw output strings. Equivalent presentations may legitimately elicit different texts, notations, and even proof strategies; what must be stable is the verdict and the obligation record. Demanding equality of raw outputs would be far too strong and is not demanded here.

Definition 6.7 (Transport equivariance)**.** A model M is *transport-equivariant* on a certified $T : x \to x'$ when, *on the joint-parse event* (18),

$$\Pi_{x'}\big(M(x')\big) \;=\; \tau_T\big(\Pi_x(M(x))\big) \qquad \text{in } A_{x'}. \tag{28}$$

Off that event the equation is not evaluated and no equivariance claim is made in either direction; non-response is charged in the response-process layer instead. For proof artifacts the corresponding condition is equality of the transported claim together with agreement of the relabelled applicable obligation coordinates (19).

Remark 6.8 (When equivariance is strictly stronger than correctness)**.** If every presentation on the orbit has a unique acceptable answer and Π returns it whenever the model is judged correct, then correctness on both x and x' already implies (28), and equivariance adds nothing. Equivariance is strictly stronger exactly when at least one of these fails: when C_x is not a singleton, so that two accepted answers need not correspond under τ_T; when Π is lossy, so that judged correctness does not determine the semantic answer; or when the artifact is a proof, whose accepted status does not identify the semantic correspondence of its content. Those are the cases the design of Section 9 is built around, and the qualification is stated here because the unqualified claim is false.

Equation (28) is the equivariance equation of geometric deep learning with the symmetry group replaced by a groupoid: where an equivariant architecture enforces $f(g \cdot x) = g \cdot f(x)$ as a structural prior, this framework *measures* the finite, certified defect of the same commutation law as a behavioural property—the finite analogue of the Lie-derivative estimator with which Gruver et al. measure learned rather than built-in equivariance in vision models [37].

6.4 The invariance defect

Definition 6.9 (Per-arrow discrepancy and the invariance defect)**.** Let d be a *separating* discrepancy on typed answers: $d(u,v) = 0$ if and only if $u = v$ (the 0–1 mismatch on final typed answers; the binary any-mismatch indicator $d_{\mathrm{bin}} \in \{0,1\}$ on the obligation vector for proof artifacts, with the Hamming severity $h \in [0,1]$ reported separately and never deciding). For a certified $T : x \to x'$ put

$$\delta_T(M) \;=\; d\big(\tau_T(\Pi_x(M(x))),\, \Pi_{x'}(M(x'))\big) \qquad \text{on } \mathrm{JP}(T), \tag{29}$$

undefined off $\mathrm{JP}(T)$ and there replaced by the composite defect of Section 9, and, for a declared sampling law ρ on certified arrows, two distinct quantities, never one:

$$D^{\mathrm{JP}}_{\mathscr{G}}(M) \;=\; \mathbb{E}_{T\sim\rho}\big[\delta_T(M) \;\big|\; \mathrm{JP}(T)\big], \qquad D^{u}(M) \;=\; \mathbb{E}_{T\sim\rho}\big[\tilde{\delta}_T(M)\big], \tag{30}$$

where $\tilde{\delta}_T$ is the composite defect of Section 9, defined everywhere. The first is a conditional mean, defined only when $\Pr[\mathrm{JP}(T)] > 0$ and reported as unavailable otherwise; it is always accompanied by that denominator, since a high conditional consistency computed on few parsed pairs is not evidence of invariance; an unconditional expectation of δ_T is not defined, because δ_T is a partial function. The second is total, and it is the one the confirmatory endpoint uses.

Remark 6.10 (Exact scope of the vanishing statement)**.** With d separating, $D^{\mathrm{JP}}_{\mathscr{G}}(M) = 0$ if and only if M is transport-equivariant ρ-almost surely on the support of ρ. Neither direction extends beyond that support: arrows outside $\mathrm{supp}(\rho)$ are unconstrained, which is Proposition 4.3 again, and the conclusion is therefore always relative to the declared sampling law. Parse failures are not silently scored as agreement or disagreement; the missingness rule of Section 9 fixes their treatment before any run.

Proposition 6.11 (Quotient non-identification)**.** *Let $T : x \to x'$ be a certified arrow with $x \neq x'$, and assume*

(a) $a_x \in C_x$ is a reference answer and $\tau_T(a_x) \in C_{x'}$ (equivariance of correctness)*;*

(b) $A_{x'} \setminus C_{x'} \neq \emptyset$ (the transformed presentation admits a wrong answer)*;*

(c) the canonical-only benchmark $\mathscr{B}$ scores each model by the verdict on x alone, i.e. $S_{\mathscr{B}}(M) = V_x(\Pi_x(M(x)))$;

(d) the augmented benchmark scores by the mean of the two binary verdicts,

$$S_{\mathscr{B}^+}(M) = \tfrac{1}{2}\Big[V_x\big(\Pi_x(M(x))\big) + V_{x'}\big(\Pi_{x'}(M(x'))\big)\Big],$$

which gives the transformed arm strictly positive weight.

Then there exist deterministic behaviours h_s, h_b with

$$S_{\mathscr{B}}(h_s) = S_{\mathscr{B}}(h_b) = 1, \qquad S_{\mathscr{B}^+}(h_s) = 1 > \tfrac{1}{2} = S_{\mathscr{B}^+}(h_b), \tag{31}$$

where h_s satisfies the transport condition (28) *on T and h_b does not.*

Proof. Let h_s return a_x on x and $\tau_T(a_x)$ on x'; by (a) both verdicts are 1 and (28) holds. Let h_b return a_x on x and, by (b), some $b \in A_{x'} \setminus C_{x'}$ on x'; then $V_{x'}(b) = 0$, so $b \neq \tau_T(a_x)$ by (a) and (28) fails. Under (c) both behaviours score 1. Under (d) the scores are 1 and $\frac{1}{2}$. □

Remark 6.12 (Why the assumptions are not decoration)**.** Each hypothesis blocks a specific counterexample to the naive statement. Without (b) every element of $A_{x'}$ is acceptable and no wrong answer exists to separate the behaviours. Without (a) the transported reference answer need not be acceptable, and the transformed arm would penalize a correct model. Without (d)—for instance if the augmented benchmark still reported only the canonical verdict, or averaged with zero weight on the new arm—adding the arm changes no score, so no separation follows from adding data alone: *what separates the behaviours is the scoring rule, not the presence of the presentation.* And when $C_{x'}$ is not a singleton, choosing b merely different from $\tau_T(a_x)$ is not enough, since such a b may itself be acceptable; this is exactly the gap that (b) closes by requiring $b \notin C_{x'}$. An earlier version of this proposition chose $b \neq \tau_T(a_x)$ without that requirement, and was therefore not proved by its argument.

Remark 6.13 (No converse)**.** The converse fails and is not claimed. A score that factors through the quotient may still be constant, or too coarse, to separate stable from brittle orbit-level behaviour: factoring through $\mathscr{X}/\sim$ constrains where the score is evaluated, not how finely it discriminates. A converse requires an additional discriminativity hypothesis—for instance that the orbit-level verdict profile $\big(V_{x''}(\Pi_{x''}(M(x'')))\big)_{x''\in[x]}$ is recoverable from the reported score—and under that hypothesis the separation is immediate. An earlier version of this proposition asserted the converse without such a hypothesis; the assertion was false.

6.5 Certificates and anti-certificates

Definition 6.14 (Certificate facets and the V0–V4 summary)**.** Every certificate records four independent facets: **(J)** justification type—assertion, argument, or derivation; **(V)** machine verification—none, sampled check, symbolic, or kernel; **(S)** tested scope—single instance, declared finite family, or all parameters used in the run; **(R)** review status—author, second expert, or external. The rungs are a summary order on facet (V) only, and they are written V0–V4 to keep them clear of the capability-claim levels L0–L4 of (1), which are a different scale entirely:

Rung	Machine verification (facet V)	Reading
V0	None	Expert assertion; contamination-prone; never confirmatory
V1	Sampled executable check	A program checks a declared invariant on a declared scope; supports admissibility, does not prove equivalence
V2	Symbolic	Computer-algebra verification of the transport identity
V3	Kernel fragment	The equivalence or its key lemma formalized in a proof assistant
V4	Kernel, end-to-end	Machine-checked equivalence

The released schema records this facet in a field named `rung` whose values are written `L0`–`L4`; the mapping is Vk = Lk, and it is stated here because the field name is frozen in the shipped bank. The rungs order machine-checkability, *not* mathematical quality: a V1 sampled check is not stronger evidence than a careful expert derivation with facet J at "derivation", merely differently supported, and the two facets must be reported side by side. In general, confirmatory admission requires V $\geq$ V2, or V1 together with S = all parameters used in the run. The *released* manifest deliberately adopts the stricter policy of an unconditional V2 floor, declining the V1-with-full-scope exception; that is a policy choice recorded in `admission.json`, not a different definition, and it is why no Module-D family is eligible.

Definition 6.15 (Anti-equivalence certificates and control typing)**.** Every negative control carries a certificate of *non*-equivalence: an executable distinguishing invariant—a cohomology class, a winding number, the value of a gauge-invariant observable—witnessing that the reference answers genuinely differ. Controls are typed:

- **N-i** (*neighbouring problem*): a well-posed problem in the same family with a valid but different reference answer. Only N-i controls are admissible in the confirmatory anti-equivalence role, because only they distinguish a degenerate normalizer from a correct one without relying on the model recognizing an impossibility.
- **N-ii** (*false-claim control*): an assertion that is simply false—a non-integral Chern number, an Euler characteristic that no closed orientable surface realizes, a boundary term declared to vanish unconditionally. These are appropriate and valuable for the obstruction track, and they are what a sycophancy instrument perturbs [5], but they are not matched non-equivalent problems and must not be used as confirmatory anti-equivalence controls.

Paradigm	Input transformation	Output relation	Certificate	Answer transport	Evaluator audit	Target construct
Metamorphic testing [36]	Domain-specific, chosen by the tester	Necessary relation between outputs	Informal or hand-proved	Possible, not standard	No	Software correctness without an oracle
Contrast sets [33], behavioural checklists [34]	Minimal human edits near a decision boundary	Label expected to change or persist	Human judgment	Label-level only	No	Local decision-boundary behaviour
Semantics-preserving perturbation, consistency evaluation [35]	Paraphrase, template or entity substitution	Output expected to be unchanged	None	Identity assumed	No	Robustness to surface variation
Template robustness for mathematics [38]	Names and numerical values inside a template	Accuracy compared across variants	Template construction	Per variant	No	Sensitivity of accuracy to instance details
Learned equivariance [37]	Continuous group action	Lie-derivative deviation from exact equivariance	Group is given	Group action	No	How far a trained model is equivariant
This paper	Certified mathematical equivalence with a declared domain	Correctness must commute with typed transport	Executable, V0–V4, with anti-certificates	Typed τ_T, reference answers transported	Yes, gated first	Transport-stable correctness (L3)

Table 2: Neighbouring paradigms. The rows differ less in ambition than in what is certified and what is audited: the transformation's validity, the answer's permitted change, and the judge's own invariance.

6.6 Relation to prior work on invariance and perturbation

Neither the observation that model correctness is fragile under content-preserving rewording, nor the idea of evaluating a system through transformations that ought not to change its output, is new here, and the contribution should be located precisely against both. The fragility phenomenon is documented systematically for grade-school mathematics by GSM-Symbolic, which varies names and numerical values within templates and reports substantial performance change [38]. The evaluation paradigm descends from *metamorphic testing* in software engineering, where necessary input–output relations stand in for an unavailable oracle—exactly the situation of a mathematical task whose correct answer is expensive to obtain but whose behaviour under a known transformation is predictable [36]. Benchmarks designed so that surface memorization is structurally unavailable are also established: PuzzLing Machines evaluates deduction on linguistic puzzles whose solution cannot be retrieved from the training distribution [32]. On the measurement side the closest relative is the geometric-deep-learning programme that treats equivariance as a quantity to be measured rather than assumed, most sharply the Lie-derivative estimator of learned equivariance [37].

What is added here is narrower than any of these paradigms and should be stated as such: transformations admitted only with a certificate and a declared domain of validity; a *typed* transport map so that the correct answer is permitted to change and is judged against its transported value rather than against string identity; correctness treated as a subfunctor rather than a single reference value; a matched *certified* surface control so that the reported quantity is an excess over rewording rather than a raw brittleness rate; and an evaluator-invariance gate that must pass before any of it is attributed to a model.

Table 2 places these side by side. The scope of the novelty claim needs care. Metamorphic testing is *defined* by input–output relations, and nothing prevents such a relation from being typed or even formally verified; typed output transport in the abstract is not the novelty. The claim made here is narrower, and it is about the audited corpus: none of the six instruments reports a consistency metric under certified semantic transport, and, within the mathematical-AI evaluation literature we surveyed—the six audited instruments, the neighbouring paradigms of Table 2, and their citing literature—we did not find the particular package assembled here: presentation-indexed answer spaces, certificate-bearing equivalences with declared domains of validity, transported correctness sets, a matched certified surface control, and an evaluator-invariance gate that must pass first. That is a statement about a bounded search, not an assertion that no such assembly exists anywhere.

7 What IMR Measures—and What It Does Not

The discipline this paper applies to the corpus applies to it reflexively, and the symmetric statement is made before any empirical claim, not after a referee asks for it.

Definition 7.1 (The target construct: transport-stable correctness)**.** Let $\mathscr{T}(x)$ be the finite set of certified arrows out of x actually administered. A model M is *invariantly correct* on the content $[x]$ when

(i) *(canonical correctness)* $\Pi_x(M(x)) \in C_x$;

(ii) *(arrow-indexed transport)* for every $T \in \mathscr{T}(x)$ the joint-parse event $\mathrm{JP}(T)$ *occurs*, and

$$\Pi_{Tx}\big(M(Tx)\big) = \tau_T\big(\Pi_x(M(x))\big) \tag{32}$$

for final-answer tasks, and, for proof tasks, equality of the transported claim together with agreement of the relabelled applicable obligation coordinates (19);

(iii) *(structural correctness)* on each administered presentation the applicable coordinates of the obligation vector are discharged.

Requiring $\mathrm{JP}(T)$ to occur, rather than merely conditioning on it, is not a technicality. A model that answered the canonical presentation correctly and then failed to produce a typed answer on every transformed presentation would satisfy a conditional-only clause *vacuously*, and would be certified invariantly correct without ever having answered a transformed problem. Non-response is charged, not excused.

Remark 7.2 (Verdict constancy is strictly weaker, and is a co-report)**.** Clause (ii) is not implied by constancy of the verdict $V_{x''}(\Pi_{x''}(M(x'')))$ across administered presentations. When $C_{x'}$ is not a singleton a model may return two individually acceptable but mutually uncorresponding answers—accepted on both arms, yet not transporting into one another—which satisfies verdict constancy and violates transport. Because the whole point of the construct is semantic correspondence rather than coincident verdicts, the definition is stated on arrows and the verdict-constancy statistic A_{orb} of (49) is reported as a weaker derived quantity, never as the construct itself. When every $C_{x''}$ is a singleton the two coincide.

Constraint 1 (Reflexive artifact non-identifiability)**.** IMR is a benchmark. Its instrument acts through an artifact projection, so Proposition 4.2 applies to it: transport stability is a property of *answers and discharged obligations*, and no measured value of it identifies an internal representation, a world model, or understanding. The point generalizes an argument of Weatherall and Wolfson, that formal correctness is neither necessary nor sufficient for the epistemic value of a proof [8]: a property certified at the level of the artifact does not thereby become a property of the process that produced it. Behavioural invariance is necessary evidence for representation-independent competence and is not sufficient for it: a lookup table indexed by both presentations of every tested orbit is transport-stable on the tested set and need not instantiate any generalizing, representation-independent reasoning process—which is Proposition 4.3 once more, now aimed at this paper.

Constraint 2 (Evaluator first)**.** No defect is attributed to a model before the evaluator's own invariance has been audited on the same orbits with expert reference answers, and no model-level claim is admissible if that audit fails its gate (Definition 9.5).

Three further limits are structural rather than incidental. *Coverage:* the construct is bounded above by problem-posing. Conjecture formation, choice of representation, selection of definitions, explanatory value, taste, interaction, and discovery are outside IMR-core; a model could be perfectly transport-stable and mathematically unremarkable [9, 10]. *Direction:* a null result is informative but not exculpatory—stability across the tested orbits would show that existing scores generalize across certified equivalence better than this analysis anticipates, and would leave every other gap of Section 4 exactly where it was. *Distinctness, not orthogonality:* quotient invariance and proof quality are logically distinct dimensions—a proof can be quotient-valid and uninsightful, or elegant and silently non-invariant—but no empirical claim of statistical independence is made here, and none would be warranted without a fitted multitrait model with held-out validation.

The honest summary is a single sentence. Certified semantic transport *targets* one of the six gaps in Section 4—it supplies evidence relevant to that gap, and closes it only if the instrument passes its gates and is actually run; the framework's own capability language must therefore stop at representation stability (L3 in (1)) and must not be extended to L4 by this paper or by any reader of it.

8 Mathematical Physics as the Stress Domain

Mathematical physics is used here not because it is difficult but because it contains naturally occurring situations in which surface form and mathematical content come apart, and in which locally flawless reasoning is globally wrong. In competition mathematics errors are usually local—a sign, a missed case, a false inequality. In the settings below every displayed line can be correct while the conclusion is false, because the argument ignored a quotient, an obstruction class, a measure, a convergence hypothesis, a boundary term, or the domain of a dictionary. Conventions are fixed throughout: mostly-minus metric $\eta = \mathrm{diag}(+,-,-,-)$, a single Abelian coupling q, Fujikawa measure-based anomalies, $\varepsilon^{0123} = +1$, and $\widetilde{F}^{\mu\nu} = \frac{1}{2}\varepsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$.

8.1 Gauge redundancy: the observable is on the quotient

Under a gauge transformation the potential changes by

$$A \longmapsto A^g = g^{-1}Ag + g^{-1}\mathrm{d}g, \tag{33}$$

so different potentials describe the same physical configuration and the physical object is the orbit $[A]$ in the quotient $\mathscr{A}/\mathscr{G}$ [23, 24]. The notation is schematic, and is used here only for the structural analogy: the action of $\mathscr{G}$ is not free at reducible connections, so the quotient is not a manifold and the appropriate object is a moduli stack or its associated groupoid; relatedly, no continuous global gauge fixing exists on the full configuration space, which is the content of the Gribov ambiguity [16] and of Singer's topological obstruction [17]. Nothing measured in this paper depends on the quotient being a manifold—what is used is that gauge-equivalent potentials are the same physical configuration, and that a quantity failing to descend to the orbit is not an observable. A quantity O is an observable only if $O(A^g) = O(A)$. This is the exact structural analogue of the evaluation problem of this paper:

$$\text{benchmark artifact} \longleftrightarrow \text{gauge representative}, \qquad \text{invariant content} \longleftrightarrow \text{gauge orbit}. \tag{34}$$

A model that computes correctly in one gauge but does not recognize that its answer is gauge-variant is competent in a chosen representative and not at the level of the quotient—and a benchmark that presents only one representative cannot tell the two apart (Proposition 6.11).

8.2 Local-to-global obstruction

The Poincaré lemma states that closed forms are locally exact on contractible sets: $\mathrm{d}\omega = 0 \Rightarrow \omega = \mathrm{d}\eta$ locally [18]. It does not state global exactness, and the obstruction is the de Rham class. The standard example must be stated with care, because the usual notation is an abuse: on S^1 there is *no* globally defined real-valued angle function θ, so "$\mathrm{d}\theta$" is not the differential of a global function. The correct object is the global angular one-form

$$\omega = \mathrm{Im}\left(\frac{\mathrm{d}z}{z}\right) = \left.\frac{x\,\mathrm{d}y - y\,\mathrm{d}x}{x^2+y^2}\right|_{S^1}, \qquad \mathrm{d}\omega = 0, \qquad \oint_{S^1}\omega = 2\pi \neq 0 = \oint_{S^1}\mathrm{d}f \tag{35}$$

for every globally defined f; hence $[\omega] \neq 0$ in $H^1_{\mathrm{dR}}(S^1)$, and ω equals $\mathrm{d}\theta$ only on charts where a branch of θ exists. A model may apply a valid local theorem repeatedly and reach a false global conclusion; the error is not algebraic but a failure to recognize an obstruction class. The Dirac monopole carries the same lesson in bundle form: patchwise potentials on S^2 with transition data of first Chern number $c_1 \in \mathbb{Z}$, whose integrality is charge quantization [19, 20]; and Gauss–Bonnet ties total curvature to the global datum χ, with $\chi(S^2) = 2$ and $\chi(T^2) = 0$.

8.3 Anomalies as measure Jacobians

A classical action may possess a symmetry, $\delta S = 0$, while the quantum theory depends also on the path-integral measure: under the transformation $\mathscr{D}\psi\,\mathscr{D}\bar{\psi} \mapsto J\,\mathscr{D}\psi\,\mathscr{D}\bar{\psi}$, and if $J \neq 1$ the symmetry is anomalous. In Fujikawa's formulation the anomaly *is* the Jacobian of the fermion measure [28]. In the fixed conventions, for a Dirac fermion of charge q in four dimensions the axial current obeys the Adler–Bell–Jackiw identity [26, 27]

$$\partial_\mu j^\mu_5 \;=\; \frac{q^2}{8\pi^2}\,F_{\mu\nu}\widetilde{F}^{\mu\nu} \;=\; \frac{q^2}{16\pi^2}\,\varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}. \tag{36}$$

Equation (36) is a local Ward identity in Minkowski signature with the conventions just fixed. The integrated statement is *not* a statement in the same setting, and running the two together is the error the module is designed to catch. Passing to the integrated form requires: Wick rotation $x^0 = -ix^4_{\mathrm{E}}$ to Euclidean signature; an orientation fixed by $\varepsilon_{1234} = +1$; and—this is the step that must be stated correctly—the identification of the relevant integer as the *Fredholm index of the chiral Dirac operator*. In Euclidean signature the Dirac operator anticommutes with γ_5 and so splits as $\not{D}_+ : S^+ \otimes E \to S^- \otimes E$ and its adjoint; the index in the Atiyah–Singer theorem [29] is

$$\mathrm{ind}(\not{D}_+) = \dim\ker\not{D}_+ - \dim\mathrm{coker}\not{D}_+ = n_+ - n_-. \tag{37}$$

The *full* operator $\not{D} = \not{D}_+ + \not{D}_+^\dagger$ is self-adjoint and, being self-adjoint on the total spinor bundle, has index zero; spectral asymmetry of a self-adjoint Dirac operator is a different invariant—the η-invariant of Atiyah–Patodi–Singer—and it is not what appears here. An earlier version of this section attributed the integer to spectral asymmetry of the self-adjoint operator; that attribution was wrong.

A second restriction is needed for the displayed formula. On a general compact spin four-manifold the index is $\mathrm{ind}(\not{D}_+) = \int_M [\widehat{A}(R)\,\mathrm{ch}(E)]_{(4)}$, which for a line bundle contributes a gravitational term as well: $\mathrm{ind} = \frac{1}{2}\int_M c_1(E)^2 - \sigma(M)/8$ with σ the signature. The gauge-only expression below is the specialization to flat $\mathbb{R}^4$ with finite-action falloff, where the configurations compactify and the $\widehat{A}$ contribution vanishes; that is the setting assumed throughout this module, and it is stated rather than left implicit. Under those conventions,

$$\mathrm{ind}(\not{D}_+) \;=\; n_+ - n_- \;=\; \frac{q^2}{32\pi^2}\int \mathrm{d}^4x\;\varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma} \qquad (\text{flat } \mathbb{R}^4,\ \text{finite action}), \tag{38}$$

integration of (36) gives the axial-charge violation

$$\boxed{\Delta Q_5 = \int \mathrm{d}^4x\, \partial_\mu j_5^\mu \;=\; 2\,\mathrm{ind}\not{D}_+)} \tag{39}$$

—*twice* the index, not the index. The factor of two is not a normalization convention but a structural fact with two equivalent readings: in the Fujikawa derivation both $\mathscr{D}\psi$ and $\mathscr{D}\bar\psi$ contribute Jacobians under $\psi \mapsto e^{i\alpha\gamma^5}\psi$, $\bar\psi \mapsto \bar\psi e^{i\alpha\gamma^5}$; equivalently, the two chiralities rotate oppositely, so each zero mode shifts the axial charge by ±1 while contributing ±1 to $n_+ - n_-$. For N_f massless Dirac flavours the same computation gives $\Delta Q_5 = 2N_f\nu$ in a background of topological charge ν [39]. The contrast worth grading is exactly this: for a *single Weyl* fermion the chiral-charge violation is $\mathrm{ind}\not{D}_+)$ itself, since only one chirality is present, and conflating the Weyl index density with the Dirac axial rotation is the canonical factor-of-two error. The non-Abelian case is a separate statement rather than a substitution into the Abelian one: for a Weyl fermion in representation R of a simple gauge group in a background of instanton number k, the index is $\mathrm{ind} = 2T(R)\,k$, with $T(R)$ the Dynkin index normalized by $\mathrm{tr}(T^aT^b) = T(R)\,\delta^{ab}$. For the fundamental of $SU(2)$, $T(R) = \frac{1}{2}$, so at $k = 1$ the index is 1 and a Dirac fermion in that background has $\Delta Q_5 = 2$.

A model that checks only the classical action will confidently "prove" quantum conservation while missing the measure contribution: quantization changes the proof obligations. A complete answer further distinguishes an anomalous *global* symmetry (physical) from a would-be anomalous *gauge* symmetry (an inconsistency that must cancel); large gauge transformations supply the discrete twin, the Chern–Simons level being quantized precisely because the winding contribution is not infinitesimally generated [22].

8.4 Regularization and limit interchange

Let $f_n = n\,\mathbf{1}_{[0,1/n]}$ on $[0,1]$. Then $f_n(x) \to 0$ for every $x > 0$—hence *almost everywhere* on $[0,1]$, but not at $x = 0$, where $f_n(0) = n \to \infty$—while $\int_0^1 f_n = 1$ for every n. Taking the a.e. limit to be the zero function, which changes nothing in the integral,

$$\lim_{n\to\infty} \int_0^1 f_n \;=\; 1 \;\neq\; 0 \;=\; \int_0^1 \lim_{n\to\infty} f_n, \tag{40}$$

and the interchange is invalid because no dominating function exists; monotone and dominated convergence and Fubini–Tonelli each fail for a stateable reason [30]. This elementary failure is the model of a ubiquitous field-theoretic hazard: formal manipulation of divergent sums, regulator-dependent expressions, contour deformations across singularities, and zero modes. The gradable obligation is not the value but the hypothesis check—which convergence theorem applies, and why.

8.5 Boundary terms

For the scalar action $S[\phi] = \frac{1}{2}\int_M |\nabla\phi|^2$, variation produces a bulk term and a boundary term,

$$\delta S = \int_M \delta\phi\,\big(-\Delta\phi\big) \;+\; \int_{\partial M} \delta\phi\,\partial_n\phi, \tag{41}$$

and the boundary contribution vanishes only under a stated condition: Dirichlet or Neumann data, compact support, closed M, suitable decay, or a compensating boundary action [24]. A derivation that discards a total derivative without checking is repeating a pattern rather than verifying a hypothesis. The graded obligation is the hypothesis check, not the ritual assertion that boundary terms vanish.

8.6 Duality—and what a pilot module can honestly claim

A duality relates two descriptions of the same physics through a dictionary that need not be an identity on variables: couplings may invert, particles may map to solitons, and observables require a nontrivial correspondence [21, 25]. The gradable question is not whether a model repeats a sentence in both descriptions but whether it transports conclusions correctly through the dictionary, within the dictionary's domain of validity.

This subsection carries a correction that also constrains the empirical design. In the pilot bank of Appendix B, the module labelled for duality is populated by families testing similarity invariance, trace and determinant identities under simultaneous change of basis, Kronecker-product determinants, and cyclicity of the trace. Those are genuine *representation-transport and naturality* tests, and they are certified at the symbolic level—but they are not physical dualities, and the module is accordingly renamed *representation and naturality transport*. No claim of duality coverage is made for the pilot; a domain-certified duality family is a prerequisite for any such claim, and is listed as a pre-run deliverable in Section 9.8.

9 A Minimal Empirical Design

This section pre-specifies the measurement protocol implied by Sections 6–8. It is written to be adoptable piecewise: the obligation record, the surface-control arm, the missingness rule, and the evaluator gate are each independently useful to any of the audited instruments.

Confirmatory identity is not a caller's choice. One property of the released implementation belongs in the specification rather than in a manual, because it is what makes the rest of this section binding. The confirmatory entry point loads the frozen manifest *internally* from its pinned location and admits no override of protocol identity: there is no argument by which a caller supplies a module set, a substitute manifest, or a subset of families. It enforces three equalities before anything is computed,

$$\mathscr{M}_{\text{input}} = \mathscr{M}_{\text{manifest}}, \qquad \mathscr{F}_{\text{administered}} = \mathscr{F}_{\text{eligible}}, \qquad \mathscr{F}_{\text{audited}} = \mathscr{F}_{\text{administered}}, \tag{42}$$

so that the module universe supplying the multiplicity denominator, the family population, and the set of orbits on which the evaluator was audited are one and the same declared object. Subsetting, custom manifests and permissive execution exist only in a separately named *exploratory* interface whose results are labelled as such and never carry an unqualified admissibility verdict. This matters because the alternative is not hypothetical: a specification that permits a caller to declare five of six modules permits a smaller Bonferroni denominator, and a specification that checks only the families it is handed permits a favourable subset.

Status of this specification. The protocol below is *pre-specified*: fixed in this document, with all constants and rules stated before any model is run. It is not, at the time of writing, *pre-registered*: that term is reserved here for a protocol deposited in a public, immutable, timestamped registry, which must precede any run whose results are reported as confirmatory. This distinction is maintained throughout the paper and is not a formality; a specification that can be edited after seeing data is not a pre-registration, whatever it is called.

9.1 The family: the minimum viable experimental unit

Definition 9.1 (Family)**.** The atomic unit is

$$\mathbf{F} = \big(x,\ T_1 x,\ T_2 x,\ s(x),\ n(x)\big)\ : \tag{43}$$

a canonical presentation x; two *certified structural* transforms $T_1, T_2 \in \mathscr{G}_{\text{str}}$, distinct administered arrows chosen to exercise different obstruction coordinates—no generating-set property is claimed or certified;

a *certified surface* control $s \in \mathscr{G}_{\text{surf}}$; and an anti-certified non-equivalent control $n(x)$ of type N-i (Definition 6.15). Reference answers for all arms, including the transported $\tau_{T_i}(a_x)$, are fixed before any model is run.

Definition 9.2 (Nested certified classes)**.** $\mathscr{G}_{\text{surf}} \subseteq \mathscr{G}_{\text{IMR}}$ is the wide subgroupoid of arrows whose certificates assert presentation-level re-expression only—wording, symbol names, notation, layout, order of given data. Such an arrow is required to be *answer-type preserving*: its certificate declares a canonical identification $\iota_s : A_x \xrightarrow{\sim} A_{s(x)}$ of the two typed answer spaces and sets $\tau_s = \iota_s$. Where the two spaces are literally the same set by construction—the usual case, since only the wording changed—$\iota_s = \mathrm{id}_{A_x}$; writing $\tau_s = \mathrm{id}$ without that declaration would be a type error, and the identification is recorded in the certificate rather than assumed. $\mathscr{G}_{\text{str}} \subseteq \mathscr{G}_{\text{IMR}}$ is the class of arrows whose certificates assert at least one structural change: chart, gauge, basis, frame, dual variables, regulator, or boundary datum. The last two require a warning that applies to the first as well. A change of regulator, of boundary data, or of duality frame is *not* automatically an equivalence: a different regulator can change the renormalized observable, different boundary data define a different boundary-value problem, and a duality dictionary outside its domain relates different theories. Such an arrow is admitted to $\mathscr{G}_{\text{str}}$ only when its certificate *proves* that the declared mathematical target—the renormalized observable, the solution functional, the graded quantity—is preserved. Where the certificate cannot establish that, the transformation is not an equivalence at all, and it is classified instead as a neighbouring problem or as a negative control (Definition 6.15), where a changed answer is the correct behaviour rather than a defect. The two classes are disjoint by declaration and *both consist of certified equivalences*. This is a correction of substance: a paraphrase that is not itself certified cannot serve as the control arm of a contrast whose treatment arm is certified, because any difference could then be attributed to the certification rather than to the structure.

9.2 Generation protocol

The analysis is only as pre-specified as the generation that feeds it, so the generation parameters are fixed here rather than left to the run.

Parameter	Fixed value
Samples per arm	$r = 4$ independent generations, identical protocol on every arm
Decoding	temperature $\vartheta = 1.0$, top-$p = 1.0$, no penalties; deterministic decoding is *not* used, so that within-arm variability is measurable
Token budget	16,384 output tokens. Truncation is classified once, and the classification is: budget-exhaustion truncation is a **parse failure** (model behaviour, composite defect), whereas a transport-layer truncation reported by the provider as an error is an **apparatus failure** (unobserved). The provider's finish reason decides, and it is recorded per response
Attempts	one attempt per generation, plus at most one format-repair re-ask quoting the required answer type and nothing else
Tools	no tool access, retrieval, or code execution in the confirmatory condition; tool-enabled conditions are a separate pre-declared secondary arm
Order	arms within a family are administered in independently randomized order across replicates, and families in randomized order
Candidate selection	none at the answer level: no best-of-n, no self-consistency decoding, no rejection sampling. The majority rule below is a *statistical aggregation of independently scored replicates*, applied after grading, not a decoding strategy
System prompt	one fixed string, identical across arms and models, shipped verbatim as `anc/prompts/system_prompt.txt`
Repair prompt	one fixed string, shipped verbatim as `anc/prompts/format_repair_prompt.txt`; it quotes the required answer type and nothing else
Seed derivation	seed = SHA256(`master`‖model‖family‖arm‖replicate) truncated to 64 bits; Python's built-in `hash` is process-randomized and is not used. Where a provider does not honour seeds, that is recorded per response and the run is reported as seed-unsupported for that model rather than treated as reproducible
Context isolation	every generation runs in a fresh conversation with no shared history, no system-prompt variation, and no caching across arms
Canonical reuse	the canonical arm is generated *independently* for each of T_1, T_2 and s; canonical replicates are never reused across arms, so that the paired differences do not share a canonical draw
Response → arm	the arm defect is the *majority* of the r replicate defects, ties resolved to defect = 1. Replicates lost to apparatus failure are *not discarded before the vote*: the arm is reduced to the interval obtained by completing the missing slots with 0 and with 1, so an arm with one observed success and three lost replicates is partially identified rather than recorded as a success, and the family and population endpoints are formed from those arm intervals
Arm → family	(50): means over observed arms within each class, then $D_{\mathbf{F}} = d_{\text{str}} - d_{\text{surf}}$
Family → module	unweighted mean of family differences within the module
Module → model	fixed pre-run weights w_m (Section 9.6)
Model → report	one confirmatory model $M^\star$; all others descriptive

Two entries deserve comment. Majority reduction of $r = 4$ replicates is deliberately coarse—it discards the within-arm rate—but it makes the family the unit of analysis without a second layer of variance components, and the discarded rates are reported descriptively so that a later design can use them. Ties at 2–2 resolve to a defect; this biases the endpoint against finding invariance failure only if surface arms tie more often than structural ones, and the tie rate of each class is reported.

9.3 The obligation record

Definition 9.3 (Structural obligation vector)**.** In place of a scalar verdict, each judged artifact receives

$$\mathbf{J}(x,y) = \big(J_{\text{logic}}, J_{\text{domain}}, J_{\text{local}}, J_{\text{global}}, J_{\text{quotient}}, J_{\text{analytic}}, J_{\text{boundary}}, J_{\text{transport}}, J_{\text{explanation}}\big) \in \{0,1,\mathsf{NA},?\}^9, \quad (44)$$

with 1 satisfied, 0 failed, NA not applicable by the item's declared applicability mask (so an irrelevant coordinate is never silently scored 0), and ? uncertain, escalated to adjudication, reported as missing and excluded from per-coordinate agreement denominators, never imputed. Each item ships a privately held annotation naming the coordinates it is engineered to stress and the specific obstruction; graders commit an initial assessment before unmasking. $J_{\text{explanation}}$ is flagged as the most fragile coordinate and is never used as a scalar proxy for understanding [10].

The vector refines the corpus instruments rather than replacing them: a binary proof label is approximately $J_{\text{logic}} \wedge J_{\text{local}}$ [4]; an executable witness check is blind to J_{quotient} and $J_{\text{transport}}$ by construction [3]; a behavioural sycophancy category coarse-grains $(J_{\text{logic}}, J_{\text{explanation}})$ on perturbed inputs [5]. The decomposition exposes which coordinates have never been measured.

9.4 Three separated estimands

Conflating the ways a transformed answer can be wrong is the single most common error in this kind of measurement, and it is avoided by definition rather than by discussion. Fix the *composite* per-arrow defect, which counts an arm-specific failure to produce a well-typed answer as a defect rather than conditioning it away:

$$\tilde{\delta}_T(M) = \begin{cases} 1, & \text{if either arm returns parse-fail after the single declared repair pass,} \\ \delta_T(M), & \text{otherwise,} \end{cases} \quad (45)$$

with δ_T as in (29). Then define, respectively, the *unconditional equivariance defect* (E1), the *defect given canonical correctness* (E2), and *transformed accuracy* (E3):

$$D^u(M) = \mathbb{E}_{T\sim\rho}\big[\tilde{\delta}_T(M)\big], \quad (46)$$

$$D^t(M) = \mathbb{E}_{T\sim\rho}\big[\tilde{\delta}_T(M) \,\big|\, \Pi_x(M(x)) \in C_x\big], \quad (47)$$

$$A_{\mathscr{G}}(M) = \mathbb{E}_{T\sim\rho}\big[\mathbf{1}\{\Pi_{Tx}(M(Tx)) \in \tau_T(C_x)\}\big], \quad (48)$$

together with orbit-robust accuracy over the declared arrow set $\mathscr{O}_i = \{\mathrm{id}, T_1, T_2\}$ of item i—the identity included, so that "every administered presentation solved" means what it says, and with behavioural non-response *charged* rather than dropped: refusal, parse failure, answer-type failure and budget truncation all give $c_{ig} = 0$, so a wholly refused arm lowers $A_{\mathscr{G}}$ and fails A_{orb} instead of vanishing from both. Apparatus failure and grader unavailability are unobserved, and they are carried as *intervals* in the co-reports exactly as in the primary endpoint: D^u, $A_{\mathscr{G}}$ and A_{orb} are each reported as a pair obtained by completing missing replicates with both outcomes, with a point value given only over *identified* arrows and labelled complete-case. An arrow with one observed success and three lost replicates is then partially identified in every quantity, rather than a definite success in the co-reports and an interval in the endpoint—

$$A_{\text{orb}}(M) = \frac{1}{N}\sum_{i=1}^{N} \min_{g\in\mathscr{O}_i} c_{ig}, \qquad c_{ig} \in \{0,1\}. \quad (49)$$

(E1) is a co-reported unconditional arrow-level defect, not the confirmatory endpoint. The confirmatory decision is taken on the lower endpoint of Δ_{paired} (Section 9.5); (E1) is the arrow-level quantity from which the family differences are built, and it is reported for interpretation rather than tested. (E2)

is the quantity that corresponds to the taxonomy label $F_{\text{transport}}$ in Table 3, which is defined only when the canonical answer is correct; (E3) is an accuracy, not a consistency. They are co-reported and never substituted for one another. Two co-reports accompany (E1): the complete-case version conditioning on joint parsing, and the directional partial-identification interval of (51), whose endpoints are formed armwise because d_{surf} enters with a minus sign [41]. Disagreement in sign or in gate status between (E1) and the lower endpoint is reported as a sensitivity flag, not resolved silently.

Denominators. For every family the denominator is the number of *administered* arrows—two structural, one surface—not the number that parsed. Families are removed from the analysis set only for a pre-run certificate revocation, never for a post-hoc parse outcome. This forecloses the selective-non-parsing failure mode, in which a model can appear more consistent by declining to answer the harder arm.

Missingness, by source. Five mechanisms produce a missing cell, and they are not interchangeable. **(i) Transport failure of the apparatus** (API error, timeout, or a truncation the provider reports as a transport error—budget-exhaustion truncation is *not* in this class, it is a parse failure, and the provider's finish reason decides which): the arm is re-administered up to three times under the identical protocol; a cell still missing is *unobserved* and enters the sharp bounds, never the composite. **(ii) Refusal**: the model produced text declining the task; this is model behaviour, is scored as a defect in the composite, and is reported separately as F_{refusal}. **(iii) Parse failure**: a well-formed attempt that Π_x cannot type after the single declared repair pass; also model behaviour, also a composite defect, reported as F_{parse}. **(iv) Judge uncertainty**: a ? coordinate in the obligation record; excluded from per-coordinate denominators, routed to adjudication, and—if adjudication does not resolve it—treated as unobserved for that coordinate only. **(v) Grader unavailable**: an item never double-graded; unobserved, entering the bounds. Categories (ii) and (iii) are behaviour and are counted as defects; (i), (iv) unresolved, and (v) are genuinely unobserved and are never imputed.

One primary estimand, over the full admitted-family population. The confirmatory target is Δ_{paired} over **every admitted family**, not over the families that happen to yield observations in both arm classes. Because apparatus failure and grader unavailability leave some arms genuinely unobserved, that target is *partially identified*: the data determine an interval, not a point. The confirmatory decision is made on the interval, and the complete-case statistic is a secondary analysis of a selected sub-population, reported under that name.

For family $\mathbf{F}$ let $S(\mathbf{F})$ and $U(\mathbf{F})$ index its administered structural and surface arrows, and let $O_\bullet(\mathbf{F})$ be the subsets whose outcome is observed in the sense above. The complete-case arm means are

$$d_{\text{str}}(\mathbf{F}) = \frac{1}{|O_S(\mathbf{F})|} \sum_{a \in O_S(\mathbf{F})} \tilde{\delta}_a, \qquad d_{\text{surf}}(\mathbf{F}) = \frac{1}{|O_U(\mathbf{F})|} \sum_{a \in O_U(\mathbf{F})} \tilde{\delta}_a, \tag{50}$$

so the denominator of each arm class is the number of *observed* arrows in that class, and a family enters the analysis only if both classes have at least one observed arrow. Refusal and parse failure never shrink these denominators—they are defects with value 1—so the selective-non-parsing route to an artificially consistent score remains closed. Apparatus failure and grader unavailability do shrink them, which is exactly why the point estimate is accompanied, always and not on request, by the sharp bounds below.

Directional partial identification, on the same population. Because d_{surf} enters with a minus sign, replacing every unobserved arm by the same constant does *not* bound the difference. The endpoints are formed *armwise, before family averaging, over all admitted families*:

$$\Delta^{-}_{\text{paired}}: \text{ unobserved structural arms} \mapsto 0, \text{ unobserved surface arms} \mapsto 1; \qquad \Delta^{+}_{\text{paired}}: \text{ the reverse}, \tag{51}$$

so that for each admitted family the realized difference satisfies $D_{\mathbf{F}} \in [D^{-}_{\mathbf{F}}, D^{+}_{\mathbf{F}}]$ whatever the unobserved arms would have shown, and hence, the weights being fixed and nonnegative, the population quantity satisfies $\Delta_{\text{paired}} \in [\Delta^{-}_{\text{paired}}, \Delta^{+}_{\text{paired}}]$ for the weighted means of the endpoints [41]. Both endpoints are

functionals of the *same* population, which is what makes them bounds; an earlier version of this specification paired a complete-case point estimate with full-population endpoints and called the latter its sharp bounds, which they were not, and separately stated the non-directional rule, which is false.

The secondary statistic targets a different population, and the mismatch is reported. The complete-case statistic is computed on the families where both arm classes yield an observation; the confirmatory endpoints cover every admitted family. The two therefore need not nest, and the complete-case value can fall outside $[\Delta^-_{\text{paired}}, \Delta^+_{\text{paired}}]$. Such a discrepancy is a *target-population mismatch*: it shows that the selected sub-population differs from the admitted one. It is consistent with informative missingness but does not identify the missingness mechanism, and it is reported as a mismatch diagnostic rather than as evidence about why the data are missing. The count of each mechanism is reported per model and per arm class, so that a reader can see whether a favourable statistic rests on an unusual amount of type-(i) attrition.

9.5 The confirmatory endpoint and its decision rule

Definition 9.4 (Structural-arm excess composite failure)**.** Within each family average the composite defect over the two certified structural arrows and subtract the matched certified surface-control defect:

$$d_{\text{str}}(\mathbf{F}) = \tfrac{1}{2}\big(\tilde{\delta}_{T_1} + \tilde{\delta}_{T_2}\big), \qquad d_{\text{surf}}(\mathbf{F}) = \tilde{\delta}_s, \qquad \boxed{\Delta_{\text{paired}}(M) = \mathbb{E}_{\mathbf{F}}\big[d_{\text{str}}(\mathbf{F}) - d_{\text{surf}}(\mathbf{F})\big].} \tag{52}$$

The name is deliberately descriptive: Δ_{paired} is the *excess composite failure of the structural arms over the matched surface arm*, and that is what it estimates without further assumptions. Reading it as the effect of structural transformation—as isolating a semantic cause—requires two assumptions that are stated here and tested rather than presumed: **(A1) matching**, that the surface and structural arms are comparable in the presentation-level features that drive brittleness for reasons unrelated to structure (length, symbol count, nesting depth, unfamiliar-token rate, position of the question); and **(A2) no differential apparatus bias**, that the evaluator and parser are not systematically less reliable on structural than on surface arms. (A1) is checked by the matching audit of threat T3 and reported as a covariate balance table; (A2) is bounded by the differential evaluator quantity (57) below. Where either fails, the estimand remains well defined and interpretable as excess failure, and the semantic reading is withheld. Causal language is reserved for a design in which arm assignment is randomized within family and the balance diagnostics pass.

Decision rule. For the single pre-specified model $M^\star$ named in the deposited protocol before any run, test

$$H_0 : \Delta_{\text{paired}}(M^\star) \le \delta_{\text{sem}} \qquad \text{against} \qquad H_1 : \Delta_{\text{paired}}(M^\star) > \delta_{\text{sem}}, \qquad \delta_{\text{sem}} = 0.05, \tag{53}$$

rejecting H_0 iff

$$\text{LCB}_{95\%}\left[\Delta^-_{\text{paired}}(M^\star)\right] > \delta_{\text{sem}}, \tag{54}$$

that is, iff a valid one-sided lower confidence bound for the *lower partial-identification endpoint* exceeds the margin, with the bound computed by the stratified cluster bootstrap of Section 9.6. Rejecting on the lower endpoint is what makes the conclusion robust to every pattern the unobserved arms could have taken. The complete-case bound is co-reported as a selected-population secondary statistic and never decides. *Gates*, all required for admissibility. Admission and apparatus validation are evaluated *before any model is run*; yield admissibility is evaluated after collection but *before any endpoint is computed*:

- **(a) Evaluator gate, with its sampling unit declared.** The audit population is the set of certified arrows in the bank, and the **independent sampling unit is the family**, not

the gold pair: the three certified arrows of one family share its mathematical content, its rendering, and its answer type, so pairs within a family are not independent draws. Two quantities are therefore reported, and they are different objects. *(a.i) Finite-bank census.* Every gold pair in the bank is audited—twenty-four families times three certified arrows, seventy-two pairs—and the observed defect is reported *exactly*, as a census proportion with no inferential interval. This describes the released instrument completely and rests on no sampling assumption. *(a.ii) Population inference.* For a claim beyond the bank, families are the clusters—and the inferential target must be declared, because the bank's families were *engineered*, not drawn at random. Two readings are available and the paper commits to reporting both.

The *census* reading of (a.i) needs no assumption at all and is the honest description of the released bank; it is the primary reported quantity.

The *superpopulation* reading requires more than the framework previously claimed. A Clopper–Pearson limit is exact for a *binomial* count, and a binomial count requires independent Bernoulli family failures with a *common* success probability p. Exchangeability alone does not deliver this: an exchangeable sequence is a mixture of i.i.d. sequences by de Finetti, and a nondegenerate mixture is over-dispersed relative to the binomial, so a Clopper–Pearson interval computed on it is not exact and can be anticonservative. The premise the gate actually needs is therefore stated as such: **(S)** the audited families are an independent probability sample from the declared target domain, and evaluator failure occurs independently across families with a common probability p. Assumption (S) is not testable on twenty-four engineered families and is not satisfied by the released bank. Consequently the superpopulation gate is *not identified for the released bank at all*, the census quantity is what the bank supports, and any future confirmatory use must either instantiate (S) by probability-sampling families from a declared frame or replace the gate with a dependence-robust or explicitly hierarchical procedure, which would be a different gate with a different critical value. Under that reading, define a *binary family-level presentation-consistency failure*

$$Z_f^{(m)} = \mathbf{1}\{\text{family } f \text{ of module } m \text{ shows any evaluator discrepancy on its audited pairs}\}, \tag{55}$$

so that $D_{\text{eval}}^{(m)} = \mathbb{E}[Z_f^{(m)}]$ is a Bernoulli mean and the exact binomial machinery applies; the continuous per-family average is reported alongside as a severity measure, but the gate is taken on Z, because an exact bound for an arbitrary bounded mean does not follow from the all-zero binomial formula.

The gate is $\max_m \text{UCB}\left[D_{\text{eval}}^{(m)}\right] < 0.05$, a maximum over $|\mathscr{M}| = 6$ modules, so the per-module bounds must be *simultaneous*. The bound is the *exact one-sided Clopper–Pearson* limit for the binomial count of failing families, at *every* count and not only at zero: a nonparametric percentile bootstrap on the same data is anticonservative in this regime—at $n = 94$, $k = 1$ and level $\alpha/6$ it returns roughly 0.043 where the exact limit is roughly 0.071, so it would pass a gate the stated procedure fails. With Bonferroni allocation each module is bounded at level $\alpha/6$, and at zero observed failures among N families *per module* the exact one-sided limit is $1-(\alpha/6)^{1/N}$. Requiring this below 0.05 gives

$$N \;\geq\; \left\lceil \ln(\alpha/6) \,/\, \ln(0.95) \right\rceil \;=\; 94 \quad \text{families per module,} \tag{56}$$

that is 564 admitted families in total. The pilot has *four* families per module, where even the marginal all-zero bound is $1-0.05^{1/4} \approx 0.527$ and the simultaneous bound is $1-(0.05/6)^{1/4} \approx 0.698$. The gate is therefore not merely unattained at pilot scale but

unattainable by more than a twenty-threefold shortfall in audit size, and the pilot is declared calibration-only rather than confirmatory. The shortfall is a factor of $94/4 = 23.5$ in families per module—more than twenty-threefold, and about 1.4 orders of magnitude. Earlier versions of this specification made this arithmetic wrong twice—first reading N as gold pairs, then pooling the bank's twenty-four families into a single bound and quoting $1 - 0.05^{1/24} \approx 0.117$ for a maximum-over-modules gate. Both are corrected here, and the corrected number is worse for the instrument, which is why it is stated plainly.

- **(b) Yield admissibility.** Require, *separately for each arm class* $c \in \{\text{structural}, \text{surface}\}$ and each model, an administered-arm response rate ≥ 0.90 and a post-repair typed-parse rate ≥ 0.80. Both denominators are the number of responses *administered*, not the number surviving the apparatus: unresolved apparatus failure lowers the response rate, since a design in which three of four replicates are lost has not achieved a 100% response rate on the one that returned. The administered, apparatus-lost, responded and parsed counts are reported per arm class so the denominator is visible rather than inferred. Failure of any of the four minima makes the primary test inadmissible for that model; a pooled rate is not sufficient, because an acceptable average can conceal a structural arm class below the floor—which is precisely the failure mode the gate exists to catch.

  Timing must be stated exactly, since an earlier version claimed all gates run before any model output is examined. That is true of the evaluator and generator validation, which use expert reference answers and no model at all, and it is what makes them pre-run. It cannot be true of yield, which is a property of model outputs. The correct sequence is: *(1)* admission filtering against the frozen manifest, and apparatus validation, both before any model is run; *(2)* collection; *(3)* yield admissibility computed *from the immutable replicate records themselves*—never from externally supplied summary rates—and recorded before any endpoint is computed; *(4)* estimation, only if every preceding gate passed. The released implementation enforces this order and refuses to compute an endpoint when a gate has failed, so that no endpoint is ever available to be inspected before its admissibility is settled.

- **(b′) Negative-control gate.** Every confirmatory family must carry a complete N-i control record—neighbouring presentation, distinct valid gold, anti-equivalence certificate, normalizer outcome, provenance digest—and that control must have been *rejected*. A missing or failed N-i control makes the run inadmissible *before* the evaluator and yield gates, because a control that was not rejected leaves a degenerate parser, normalizer or transport un-excluded, and every downstream quantity would then be uninterpretable.

- **(c) Excision, and what it does to the estimand.** If a module fails (a), the pre-declared action is that *the run is inadmissible as confirmatory*. Module excision is permitted only as a labelled secondary analysis, and when performed it changes the target population: the estimand becomes Δ_{paired} over the sub-population of modules passing the gate, is reported under that name, and is never presented as the original endpoint. Silent retargeting is the failure mode this clause exists to prevent.

Co-reports never substitute: D^t, $A_{\mathscr{G}}$, A_{orb}, the per-coordinate $\mathbf{J}$ profiles, and the typed failure rates are reported alongside, and no co-report rescues a failed primary test or a failed gate.

This is a superiority-over-margin test, and it is stated as such because the earlier formulation was not: a rule of the form LCB $> \delta$ does not test the null "no excess" but the practical-margin null (53), and reporting it as the former overstates what rejection means [42].

Justification of the margin. $\delta_{\text{sem}} = 0.05$ is the declared smallest effect size of interest, and it is justified by decision consequences rather than by coincidence with any other constant in the design.

The anchor is stated on the scale of the paired difference itself, because Δ_{paired} is not a defect rate: it is the difference between a mean of two structural defects and a matched surface defect, and a surface failure offsets a structural one. The reporting consequence is therefore this. Suppose an instrument reports a canonical-presentation score and a reader wishes to treat it as informative about the content rather than the presentation. If structural reformulation costs no more than five percentage points of correctness beyond what mere rewording already costs, the canonical score remains a usable proxy and the practical reading of published leaderboards is unchanged; above that, the ordering a reader takes from a canonical score can invert under a reformulation that a mathematician would regard as the same problem, and the score must be reported with an invariance qualifier. One further scale caveat is needed, because δ_{sem} does not live on the scale of ordinary response correctness. With $r = 4$ replicates reduced by a conservative majority (Section 9.2), the arm defect is a nonlinear function of the replicate defect rate, and five points of majority-reduced paired defect correspond to a smaller change in mean replicate correctness near the middle of the range and a larger one near its ends. The margin is therefore declared *on the majority-reduced paired scale*, which is the scale the endpoint is computed on, and the simulation grid of Section 9.6 reports size and power on that same scale—so the choice is calibrated against the estimand it applies to rather than translated from an arrow-count intuition that does not describe it.

It would be a mistake—and an earlier version of this design made it—to justify δ_{sem} by pointing at the evaluator gate constant merely because both are 0.05. The gate bounds a *marginal* evaluator defect rate; what threatens the paired contrast is the *differential* quantity

$$\Delta_{\text{eval}} = \mathbb{E}\big[d_E \text{ on structural arms}\big] - \mathbb{E}\big[d_E \text{ on surface arms}\big], \tag{57}$$

which may be larger than either marginal rate, may cancel, or may carry the opposite sign. Δ_{eval} is estimated on the same gold-pair audit as D_{eval}, reported with a cluster-bootstrap interval, and subtracted in a pre-declared sensitivity analysis of the primary endpoint. The two constants are independent design choices that happen to take the same value.

The threshold sensitivity object is *engine-dependent*, and both forms are declared here so that neither is mistaken for the other: when the percentile-t bound decides, the reported curve is $\delta \mapsto \text{LCB}_{95\%}[\Delta^{-}_{\text{paired}}] - \delta$, positive exactly where the margin would be rejected; when the fallback decides, no studentized lower bound is available and the reported object is instead $\delta \mapsto p_{\text{wild}}(\delta)$, the null-imposed wild-cluster p-value at each margin, which rejects where it falls below 0.05. The released implementation returns the object belonging to the deciding engine and labels which one it is. The appropriate curve is reported so that any reader may read off the decision at any margin, together with pre-declared sensitivity analyses at $\delta \in \{0, 0.02, 0.10\}$; the point-null value $\delta = 0$ is reported explicitly and labelled as a different, weaker hypothesis.

Multiplicity. The confirmatory test applies to $M^{\star}$ alone. Across a model set the pre-specified global test is $H_0^{\text{glob}} : \max_M \Delta_{\text{paired}}(M) \le \delta_{\text{sem}}$, evaluated by the studentized-maximum cluster bootstrap with familywise error control; every other model-wise interval is descriptive and is labelled as such in every table. No claim of a multiplicity-free single endpoint is made when several models are run—an inconsistency in the earlier specification that this rule removes.

9.6 Analysis specification

Responses are not independent: several arms come from one family, several graders judge one response, several samples come from one model, and families belong to modules. Binomial intervals are invalid; the analysis is clustered, and the algorithm is fixed in advance.

Estimator and variance. Let module m contain N_m admitted families and let $D_{\mathbf{F}} = d_{\text{str}}(\mathbf{F}) - d_{\text{surf}}(\mathbf{F})$ be the family-level difference. The point estimate is the stratum-weighted mean $\hat{\Delta} = \sum_m w_m \bar{D}_m$, where $\bar{D}_m$ averages the observed family differences in module m and the weights are **fixed before the run** from the *admitted* family counts, $w_m = N_m^{\text{adm}} / \sum_{m'} N_{m'}^{\text{adm}}$ (equal weights when the bank is balanced, as in

the pilot). Weights are never renormalized over the modules that happen to yield observations: doing so makes them post-outcome quantities and silently retargets the estimand toward modules with better apparatus yield. If a module contributes *no* observed family the run is declared inadmissible rather than pooled or recoded; a reweighted analysis over the surviving modules may be reported, but only under a different name, as a retargeted estimand. The family is the influence unit, and the variance estimator is

$$\widehat{\mathrm{Var}}(\hat{\Delta}) = \sum_m \frac{w_m^2}{N_m(N_m-1)} \sum_{\mathbf{F}\in m} \big(D_{\mathbf{F}} - \bar{D}_m\big)^2. \tag{58}$$

Resampling counts, declared separately. The confirmatory bound uses $B_{\mathrm{prim}} = 20{,}000$. The pre-declared secondary procedures use their own fixed counts, stated here so that code and protocol agree: $B_{\mathrm{wild}} = 4{,}000$ for the fallback p-value, $B_{\mathrm{curve}} = 4{,}000$ for the threshold curve, $B_{\mathrm{sens}} = 2{,}000$ for the differential-evaluator sensitivity. These are fixed, not adaptive. Moreover the threshold curve and the sensitivity analysis are computed with *the engine that decided the primary test*: if the fallback rule selected the wild bootstrap, they are recomputed under it, since reporting a sensitivity analysis from a method already declared unreliable for that dataset would be incoherent.

Resampling algorithm. (1) For $b = 1,\dots,B_{\mathrm{prim}}$ with master seed 20240517: within each module stratum draw N_m families with replacement; recompute $\hat{\Delta}^{(b)}$ and $\widehat{\mathrm{Var}}^{(b)}$ by (58) on the resample. (2) Form the studentized statistic $t^{(b)} = (\hat{\Delta}^{(b)} - \hat{\Delta})/\sqrt{\widehat{\mathrm{Var}}^{(b)}}$. (3) Report the one-sided percentile-t lower bound $\mathrm{LCB}_{95\%} = \hat{\Delta} - q_{0.95}\{t^{(b)}\} \cdot \sqrt{\widehat{\mathrm{Var}}(\hat{\Delta})}$. (4) *Zero-variance handling*: if $\widehat{\mathrm{Var}}^{(b)} = 0$ for a resample—possible with binary defects and few clusters—that resample is retained and its $t^{(b)}$ is defined as 0 if the numerator is also zero and as $\pm\infty$ otherwise, with the proportion of infinite values reported; if more than 5% of resamples are degenerate the percentile-t bound is declared unreliable and the fallback is used.

Fallback. Whenever any stratum contributes fewer than eight usable families, or the degeneracy proportion exceeds 5%, the pre-declared fallback is the *null-imposed* wild cluster bootstrap with Rademacher weights [40]. Impose H_0 by a *constrained stratified fit* rather than by global recentring, so that module heterogeneity survives the null. Let $\bar{D}_m$ be the observed mean of module m and $\hat{\Delta}$ the stratum-weighted estimate. Set the fitted values and residuals

$$\hat{D}^0_{\mathbf{F}} = \bar{D}_{m(\mathbf{F})} + \big(\delta_{\mathrm{sem}} - \hat{\Delta}\big), \qquad \hat{e}_{\mathbf{F}} = D_{\mathbf{F}} - \hat{D}^0_{\mathbf{F}}, \tag{59}$$

so that every module mean is shifted by the same constant and the weighted mean of the fitted values equals δ_{sem} exactly, while the between-module contrasts are preserved. Global recentring, $\tilde{D}_{\mathbf{F}} = D_{\mathbf{F}} - \hat{\Delta} + \delta_{\mathrm{sem}}$, would impose the same weighted mean but is a different null fit and yields a different bootstrap distribution; (59) is what the released implementation performs, and since the fallback is decision-bearing in sparse strata the distinction is not cosmetic. Resample by multiplying $\hat{e}_{\mathbf{F}}$ by an independent Rademacher draw shared across that family's arms; recompute the studentized statistic; and take the one-sided p-value as $(1+\#\{\text{valid replicates at least as extreme}\})/(1+\#\{\text{valid replicates}\})$. Degenerate replicates, those with zero resampled variance, are excluded from *both* the numerator and the denominator and their count is reported; retaining $B_{\mathrm{wild}}+1$ in the denominator while discarding degenerate draws would understate the p-value, and in sparse binary designs such draws are not rare. If any stratum contributes no usable family the run is declared inadmissible rather than pooled.

Multiplicity statistic. For the global test the statistic is the studentized maximum $T_{\max} = \max_M \big(\hat{\Delta}_M - \delta_{\mathrm{sem}}\big)/\sqrt{\widehat{\mathrm{Var}}(\hat{\Delta}_M)}$, with its null distribution obtained by the same family-level resampling applied *jointly* across models—families resampled once per replicate and all models recomputed on that resample, preserving the within-family correlation of model outcomes. Joint resampling requires a *common support*: the models must be compared on one aligned family vector. The support is fixed pre-run as the set of admitted families administered to every model in the declared set, and the partial-identification endpoints are formed on that same common support. Where a model's missingness pattern differs, the global test

is run on the common panel and the resulting population is labelled as such; a per-model complete-case support would make $T_{\max}$ a maximum over quantities defined on different populations, which is not a test statistic.

Balance. Equal family counts per module are a *design choice* that simplifies interpretation of per-module contributions. They are not required by a stratified bootstrap, and four clusters per stratum is sparse—which is precisely why the fallback above and the calibration below are pre-declared, and why scale-up raises the per-module cluster count rather than the number of modules.

Calibration, with the generating process written down. The procedure is calibrated before the run on a simulation grid, not a single process. The generating process reproduces the *administered* protocol of Section 9.2, not a simplified version of it. Family i in module m carries a latent brittleness $u_{im} \sim \mathcal{N}(0, \sigma_m^2)$ shared by its arms, inducing within-family correlation. Each arm is generated $r = 4$ times; each replicate defect is Bernoulli with $\operatorname{logit} p = \mu_m + u_{im} + \beta \mathbf{1}\{\text{structural}\}$, and the arm defect is the *conservative majority* of the four replicates, ties resolving to a defect—so the simulated endpoint is the majority-reduced quantity the experiment actually computes, not the mean replicate defect. The three response-process mechanisms are kept distinct, matching Section 9: apparatus failure with probability π_{app}, optionally offset on structural arms, yielding an *unobserved* arm; refusal with probability π_{ref} and parse failure with probability π_{parse}, each yielding a *composite defect*; and judge error flipping an observed replicate defect with probability π_J. Because majority reduction is nonlinear, β is calibrated by inverting the realized majority-reduced excess rather than the replicate-level one. The calibrated target is the realized *directional lower endpoint* $\mathbb{E}[\Delta^-_{\text{paired}}]$, not a latent excess: β is solved per cell by bisection against that quantity under the complete paired response process, with calibration replications, tolerance 0.006 and seed policy recorded in every output row. The grid is $\mathbb{E}[\Delta^-_{\text{paired}}] \in \{0, 0.025, 0.05, 0.075, 0.10, 0.15\}$, including the zero target, which is solved rather than assumed to be $\beta = 0$; module heterogeneity σ_m and μ_m at ratios 1:1, 2:1, 4:1; $\pi_{\text{miss}} \in \{0, 0.05, 0.15\}$ with and without an arm-dependent offset; $\pi_J \in \{0, 0.02, 0.05\}$; $N \in \{24, 30, 48\}$; 2,000 replicates per cell. Each simulated dataset is generated as *paired* canonical and transformed responses through the same record type, the same certified transport, the same replicate-to-arrow majority and the same directional bounding that the experiment uses, and the endpoint, the percentile-t bound and the fallback selection are computed from those records. The scope is stated exactly rather than overclaimed: the grid calibrates the *inference engine on paired records*. It does *not* exercise the fail-closed confirmatory gates—manifest identity, the audit provenance join, and the N-i control—which are exercised instead by the released regression suite against a fully valid synthetic registry. Reported per cell: empirical rejection rate at the null boundary $\Delta_{\text{paired}} = \delta_{\text{sem}}$ (the size), power at alternatives strictly above it, the degeneracy proportion of the percentile-t bound, and the frequency with which the fallback engine decided. An earlier version calibrated the percentile-t bound alone while reporting a fallback frequency of one, so that the tabulated rejection rates belonged to a procedure the primary analysis would never have used. A single coverage figure at one configuration—the earlier specification reported 0.949 ± 0.005 for one synthetic process—calibrates one point of this grid and no other, and is not evidence of size control at the boundary.

Power. Power is targeted at $\Delta_{\text{paired}} = 0.15$, three times the margin, not at δ_{sem}: at the boundary power equals size by construction, so a power calculation aimed there is uninformative. The pilot at $N = 24$ is a *pipeline and calibration study*, not a powered one: it validates transformations, grader agreement, coordinate reliability, and the analysis path end to end, and it cannot validate evaluator invariance at all, since its evaluator gate is mathematically unattainable at four families per module (56). Both the confirmatory family count and the audit size are set by the simulation grid and by (56), and fixed in the deposited protocol before any confirmatory run.

Annotation model, matched to the four-valued record. This subsection is an analysis plan, not a released procedure: no implementation of it accompanies this paper, and no result reported here depends on it. The obligation vector takes values in $\{0, 1, \mathsf{NA}, ?\}$, so a latent-truth model cannot be

Type	Meaning	Unit and role
F_{refusal}	Model declines or makes no attempt	Response; yield gate
F_{parse}	Π_x returns parse-fail: nothing extractable	Response; composite defect
$F_{\text{answer-type}}$	Extractable but wrongly typed for the declared answer type	Response; composite defect
$F_{\text{canonical}}$	Wrong on the canonical presentation itself	Response; not a transport failure
$F_{\text{transport}}$	Correct on x, not equivariant on Tx	Response; corresponds to (E2)
$F_{\text{mechanism}}$	Right answer, unsound mechanism	Response; surfaced by **J**
F_{witness}	Claims existence without the required witness	Response; surfaced by **J**
$F_{\text{obstruction-class}}$	Names the wrong structural obstruction	*Obligation record*, not the router
$F_{\text{evaluator}}$	Defect localized to the judge	Item/system; set by the gate
$F_{\text{generator}}$	Defect localized to the item or transform	Item/system; set by validation

Table 3: The attribution taxonomy. It is *descriptive*: it routes observed outputs to a reporting channel and does not by itself identify causes. Response-level labels are assigned by the deterministic precedence $F_{\text{refusal}} \succ F_{\text{parse}} \succ F_{\text{answer-type}} \succ F_{\text{canonical}} \succ F_{\text{transport}} \succ F_{\text{mechanism}} \succ F_{\text{witness}}$, which is the order implemented by the released router. Answer-typing precedes canonical correctness of necessity: canonical equality is undefined on a wrongly typed output—one cannot ask whether a returned matrix equals a scalar reference value. $F_{\text{obstruction-class}}$ is *not* a router label; it is a judgment recorded in the obligation vector (coordinates J_{global}, J_{quotient}, J_{boundary}), and it is listed here so that the two senses of "type" are never conflated. The two system-level labels are assigned by the gates and override response-level labels for the affected items. Causal language is reserved for claims backed by intervention or replication.

fitted to it as a single categorical variable. The pre-declared scheme is: for each of the nine coordinates, restrict to the records where the coordinate is *applicable* on the item's declared mask (dropping NA), treat ? as *missing at random given the item*, and fit a binary Dawid–Skene model on the remaining $\{0,1\}$ labels with two independent graders per record and a third adjudicating grader invoked for every record with a ? or a disagreement [31]. Priors: symmetric $\text{Beta}(2,2)$ on each grader's per-class accuracy and $\text{Beta}(1,1)$ on class prevalence, with accuracies constrained above 0.5 to fix the label-switching mode. The assumptions are stated because they are not innocuous here: a single latent class per record and conditional independence of graders given the truth may both fail for structural judgments, where graders share training and can share a misreading. Pre-declared diagnostics are grader-pair residual correlation and item-difficulty covariate checks; the pre-declared alternative model, reported alongside whenever the residual correlation exceeds 0.2, is a hierarchical logistic model with item and grader random effects and a grader-pair interaction. Sensitivity to plain majority vote is reported in every case, and the ?-rate is reported per coordinate as adjudication load.

9.7 Attribution taxonomy and the evaluator gate

Definition 9.5 (Evaluator presentation-consistency defect)**.** Let L_x denote the judge's output space on presentation x: for final-answer tasks $L_x = \{0,1\}$, and for proof tasks $L_x = \{0,1,\text{NA},?\}^9$, the obligation record. Let d_E be the discrepancy on L_x: absolute difference of verdicts in the first case; in the second, the *binary any-mismatch indicator* $d_{\text{bin}} \in \{0,1\}$: one if any coordinate comparable on both sides differs, zero otherwise, with NA-on-both excluded, ? on either side routed to adjudication and excluded, and an applicability mismatch counted as a discrepancy. The per-coordinate Hamming fraction over the comparable set, written $h \in [0,1]$, is reported alongside as a severity summary and is never the decision.

The two symbols are distinct throughout, in paper and code: d_{bin} decides, h describes. In both cases d_E is reduced to one value per family. Two interval statements are then made, and they are different objects. The *gate* is taken on the binary family-level failure indicator and uses the *exact* one-sided Clopper–Pearson limit, Bonferroni-corrected across the frozen module universe; the continuous per-family average is reported as a *severity* summary with a family-level nonparametric cluster bootstrap interval, and never decides the gate. No parametric observation model is imposed on the obligation record: specifying one would require the within-proof correlation structure across nine coordinates, which is not identified at pilot scale. What this audits is *presentation consistency*, not evaluator accuracy: it asks whether the judge treats a reference answer and its transported counterpart alike, and a judge that is consistently wrong on both would pass. Accuracy calibration against gold labels is a separate quantity, reported separately and not gated here. Before any defect is attributed to a model, the judge E is tested on expert-written reference answers transported across each orbit:

$$D_{\text{eval}} = \mathbb{E}_{T\sim\rho}\Big[d_E\big(E(x,a_x),\ E(Tx,\tau_T(a_x))\big)\Big]. \tag{60}$$

If the judge accepts a reference answer in one presentation and rejects the transported one in another—on account of notation, ordering, proof style, basis, or representation—the discrepancy is the evaluator's, is classified $F_{\text{evaluator}}$, and is removed before any model-level claim. The audit set includes notation changes, reordered arguments, alternative valid proof styles, coordinate and basis changes, and equivalent formal statements; LLM judges are additionally probed for verbosity, positional, self-preference, and familiarity biases.

Two properties of this gate must be stated precisely, because overstating them would repeat the error this paper criticizes. First, for a deterministic symbolic judge that decides equality through the same kernel that defines the reference answers, $D_{\text{eval}} = 0$ is *kernel self-consistency calibration*—a check that the apparatus does not contradict itself—and not evidence about judging behaviour. Second, the renderer, the parser, and any live judge must be validated separately, on expert-equivalent artifacts rather than on rendered reference values; no live-judge D_{eval} is reported in this paper because none has been measured, and the battery, comparison procedure, and gate constant are fixed here so that the measurement cannot be tuned afterwards.

9.8 Hypotheses and threats to validity

The design decides falsifiable hypotheses, stated with the estimand that decides each.

Hypothesis 1 (Structural excess)**.** $\Delta_{\text{paired}}(M^\star) > \delta_{\text{sem}}$*: composite defect under certified structural transformation exceeds the matched certified surface-control defect by more than the margin. Decided by* (53).

Hypothesis 2 (Rank discordance)**.** *There exist models with* $S_{\text{std}}(M_a) > S_{\text{std}}(M_b)$ *yet* $A_{\mathscr{G}}(M_a) < A_{\mathscr{G}}(M_b)$. Status: descriptive, with no decision rule. *No test is defined and none is claimed.* S_{std} *comes from a different instrument with its own sample, protocol, and uncertainty structure, so a family-level bootstrap over the IMR bank cannot furnish a joint band for a rank difference across the two; supplying one would require a common dataset, a declared ranking functional, a tie rule, a joint resampling law, and a familywise procedure, none of which the released design has. Rank discordance is therefore reported as an observation—the two orderings side by side, with per-model intervals on* $A_{\mathscr{G}}$ *alone—and any inferential reading is deferred.*

Hypothesis 3 (Structural exceeds parametric)**.** *Certified structural transformation induces larger defect than parameter variation within a fixed presentation [3, 38].* Status: exploratory, outside the released instrument. *Testing it requires an extended family* $\mathbf{F}^+ = (\mathbf{F}, p(x))$ *carrying a parameter-variation arm,*

together with a certificate stating how the parameter is drawn, how its reference answer is computed, and how it is difficulty-matched to the structural arms. The released bank has no such arm and the schema does not define one, so $\mathbf{F}^+$ *is specified here as a design target rather than exercised. Until it exists, any comparison against parameter-variation studies [3, 38] is a cross-study observation and is labelled as one.*

Hypothesis 4 (Judge presentation-sensitivity)**.** *Live judges grade certified-equivalent correct artifacts differently on account of surface form.* Decision rule: *declared satisfied when* $\mathrm{LCB}_{95\%}\left[D_{\mathrm{eval}}^{\mathrm{live}}\right] > 0$ *on the gold-pair audit, with* D_{eval} *as in Definition 9.5 and the interval from the same family-level resampling. This is a hypothesis* about the apparatus*, and it does not license a model-level claim. Nor does satisfying it automatically disqualify one: the admissibility gate tolerates a presentation-consistency defect provably below* 0.05*, so a defect that is reliably positive but bounded below the gate leaves the primary test admissible while establishing H4. Detecting a nonzero judge defect and tolerating a small bounded one are different findings, and an earlier version of this specification conflated them.*

Four threats are stated openly, in decreasing severity, so that no reader learns them first from a referee.

(T1) Ceiling. The pilot's canonical problems are deliberately textbook-level, so that failure cannot be excused as computational difficulty; the price is a possible ceiling effect for frontier models, in which case the pilot's information about Hypothesis 1 is limited and scale-up must raise canonical difficulty while preserving certifiability.

(T2) Transport triviality. In the current bank all but one family have identity answer transport, the invariant being numerically unchanged across the orbit. The groupoid machinery—and with it the loop condition (16), which is vacuous when $\tau = \mathrm{id}$—is therefore nearly unexercised empirically, and the typed-transport contribution cannot be evidenced at this scale. Two consequences are pre-declared: a *non-identity stratum estimand*, namely Δ_{paired} restricted to families with $\tau_T \neq \mathrm{id}$, reported with its own interval; and a requirement that no claim about typed transport be made until that stratum contains at least six families across at least three modules. Flagship non-identity families—non-Abelian holonomy under gauge conjugation with transported character data, de Rham period obstructions with re-coordinatized reference answers, certified limit-interchange failures under regulator change—are designed and symbolically verified but not yet in the bank.

(T3) Surface-control matching. The surface control is certified and difficulty-matched by construction protocol, but not yet *complexity-matched* to the structural arms in measured surface statistics (length, symbol count, nesting depth, unfamiliar-token rate). Since Δ_{paired} is a difference, an unmatched control biases the confirmatory endpoint in an unknown direction; a matching audit reporting those statistics for both arms is a pre-run deliverable.

(T4) Module heterogeneity in certificate strength, and the six-module deadlock. There is an operational consequence worth stating plainly, because it binds the whole programme: the evaluator gate is a maximum over the frozen *six*-module universe, so a run in which Module D is excluded cannot be confirmatory—it would take the maximum over five and use a Bonferroni denominator the protocol does not authorize. The manifest therefore records the exact transition required: Module D becomes eligible only when all four of its families carry V2 certificates replacing the V1 ledger, *and* carry type N-i negative controls, *and* carry complexity-matched surface controls. Until that transition, no confirmatory analysis is permitted at all, and the released manifest enforces this by recording every family as ineligible. Module D currently carries V1 convention-ledger certificates: the stated normalization is machine-checked for algebraic consistency, not derived from first principles. Until promoted, Module D is excluded from the confirmatory estimand and reported only in the exploratory layer.

Level	What is established	Evidence required to reach it
L0	Artifact acceptance: an evaluator accepted a number, witness, proof text, label, or term	The run itself, with the protocol and judge specified
L1	Task performance on a declared protocol and distribution	Reliability, contamination control, adequate sample, reported protocol
L2	Generalization to a declared target population	An explicit target definition, a coverage audit, and out-of-sample transfer evidence
L3	Representation-stable correctness	*Definition*: arrow-indexed transport compatibility (Definition 7.1). *Evidence*: certified arms judged against transported reference answers, a matched surface control, a passed evaluator gate
L4	Broad mathematical capability	*Would require at minimum*: a measurement model whose dimensionality, local-independence, invariance, and DIF assumptions are tested; held-out predictive validation; and coverage of posing, explanation, and interaction. Necessary, not jointly sufficient

Table 4: The claim matrix of (1): five *evidence dimensions*, not a ladder. The ordering is expository, *not* logical: the levels are not nested, and in particular L3 does not imply L2—a model can be transport-stable on a finite certified orbit and fail on a fresh problem from the target population, and conversely. Each row states a distinct evidence requirement that the others do not supply. The corpus audited here establishes L0–L1 robustly and reaches toward L2; this paper's framework targets L3; L4 is claimed by no instrument, including this one.

10 Implications for Benchmark Reporting

The critique is actionable, and this section states the actions. Nothing here requires abandoning an existing instrument; each item is a reporting or design change that any of the audited projects could adopt without redesigning its pipeline.

10.1 A claim matrix

The recommendation is simple: *report at the level the evidence supports, and name the level*. "Model M solved 71% of the 2026 competition problems under protocol Γ, graded by J" is an L1 statement and is fully supported. "Model M has strong mathematical reasoning capability" is an L4 statement and is supported by no instrument in the corpus. Naming the level is the whole discipline; the levels do not stack, so reaching one says nothing about another. The distance between those two sentences is the entire subject of this paper.

10.2 Six reporting items

1. **Name the estimand and the protocol in the headline claim.** A score is a property of a compound system (3); reporting it as a property of the model alone is an incomplete attribution—protocol-relative shorthand presented as a property of the system under test—and it costs nothing to avoid.
2. **Publish a coverage audit.** For the declared target domain, report the fraction of an expert-sampled task list admitting a feasible evaluator under stated budgets, and name the excluded strata (Section 4.3). This converts verifier-selection bias from a criticism into a measured quantity.

3. **Audit the judge for presentation sensitivity.** Run the evaluator on equivalent rewritings of accepted artifacts and report its own defect before attributing any difference to models (Definition 9.5). For LLM judges, report the latent-annotator fit, not only raw agreement.
4. **Replace, or supplement, the scalar verdict.** A structured obligation record separates failure modes that a single bit merges (Definition 9.3), and it is what makes a benchmark diagnostic rather than merely comparative.
5. **Add a certified-equivalence arm with a certified surface control.** The minimum is *three administered presentations per item*: the canonical one, at least one certified structural transform judged against its transported reference answer, and one certified surface control with the same answer type; a fourth, an anti-certified non-equivalent control, is needed before the comparison can be trusted. Report the structural-minus-surface difference, not the raw structural defect. This is the smallest intervention that turns an L1 instrument into one that speaks to L3.
6. **If a latent-trait model is fitted, report its diagnostics.** An explicit item-response model is better practice than an implicit weighting, and the platform's is a model of the right kind [2]. What should accompany it are the checks its interpretation requires—dimensionality, local independence, cross-tier measurement invariance, differential item functioning, missingness sensitivity, and held-out predictive validation—or else the summary should be labelled a composite index with declared weights rather than a capability estimate (Section 4.5).

11 Conclusion

The audited benchmarks yield valuable measurements of performance on explicitly defined mathematical tasks under explicitly defined protocols. The criticism advanced here concerns the interpretation of those measurements. Artifact acceptance, task accuracy, judge agreement, contamination resistance, and formal kernel acceptance do not by themselves identify a general, representation-independent mathematical-reasoning capability: the score-to-capability step is an identification problem (5) that no published design in this corpus solves, and it is left open by six specific and separately addressable gaps—artifact projection, finite-sample non-identification, verifier-selection of the task distribution, judge substitution, the untested measurement assumptions behind a scalar capability index, and formalization adequacy.

Certified semantic transport supplies one missing validity test, and one only: whether accepted correctness is stable across rigorously established reformulations of the same mathematics. The apparatus for that test is a groupoid of invertible certified equivalences, an answer functor into sets and bijections, an equivariant correctness subfunctor, and a measurement protocol with separated estimands, a declared missingness rule, a margin-based decision, a multiplicity policy, and an evaluator gate that must pass before any model-level claim. Passing that test would strengthen a capability claim; failing it would delimit the scope of existing scores; and either outcome is reportable under a rule fixed before the data exist.

A tensor is not its coordinate array; a gauge field is not one potential; a cohomology class is not one cocycle; a theorem is not one string of symbols. Whether the mathematical behaviour of large language models descends from presentations to equivalence classes is a precisely posed empirical question with a specified instrument and an unrun experiment. What it is not, and what this paper has been careful never to assert, is a question already answered in either direction.

Artifact availability

The reference implementation of the measurement core, the materialized family artifacts, the executable certificates, the verification and test suites, and the manifest described in Appendix B accompany this paper as ancillary files. No pre-registration is claimed: the protocol of Section 9 is pre-specified in this document, and any confirmatory run must be preceded by deposit of that protocol in a public, immutable, timestamped registry.

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A Notation

Symbol	Meaning
$\mathscr{B} = (D_{\mathscr{B}}, A_{\mathscr{B}}, J_{\mathscr{B}}, h_{\mathscr{B}}, \Gamma_{\mathscr{B}}, R_{\Gamma_{\mathscr{B}}})$	Benchmark instrument: records, artifacts, judge, scoring map, protocol, renderer
$\pi_{\mathscr{B}}$	Artifact projection $\mathscr{R} \to \mathscr{A}_{\mathscr{B}}$
$\Theta_{\text{math}},\ \eta_M$	Capability parameter; nuisance response processes in (5)
$\mathscr{P}_{\text{eval}},\ \mathscr{P}_{\text{target}}$	Feasibly evaluable problems; declared target domain
$\mathscr{G}_{\text{IMR}},\ \mathscr{G}_{\text{str}},\ \mathscr{G}_{\text{surf}}$	Certified-equivalence groupoid; structural and surface subclasses
$A : \mathscr{G}_{\text{IMR}} \to \mathbf{Set}_{\cong}$	Answer functor; A_x the typed answer space of presentation x
$\tau_T = A(T)$	Typed answer transport along the certified arrow T
$C_x \subseteq A_x$	Correctness subfunctor; $\tau_T(C_x) = C_{Tx}$; reference answer a_x when $\lvert C_x\rvert = 1$
Π_x	Semantic normalizer, valued in $A_x \sqcup \{\mathsf{parse\text{-}fail}\}$
V_x	Verdict map; quotient validity is $V_{Tx} \circ \tau_T = V_x$
$\delta_T,\ \tilde{\delta}_T$	Per-arrow discrepancy; composite defect counting non-response
$D^{\mu}, D^{t}, A_{\mathscr{G}}, A_{\text{orb}}$	Unconditional arrow-level defect (co-report); canonical-conditioned defect; transformed accuracy; orbit-robust accuracy
$\Delta^{-}_{\text{paired}}$	Lower partial-identification endpoint; *the decision-bearing quantity*
Δ_{paired}	Structural-arm excess composite failure: structural minus certified-surface defect (Definition 9.4)
D_{eval}	Evaluator *presentation-consistency* defect on transported reference answers (Definition 9.5); not an accuracy measure
$\mathbf{J} \in \{0, 1, \mathsf{NA}, ?\}^9$	Structural obligation vector
$\delta_{\text{sem}} = 0.05$	Pre-specified margin (smallest effect size of interest)

B The Pilot Family Bank and Released Artifacts

The design of Section 9 is *partially* instantiated in a calibration bank of twenty-four certified families, four in each of six obstruction modules. The bank is not a confirmatory instantiation of Definition 9.1, and the fields that fail confirmatory admission are these: the negative controls are predominantly type N-ii rather than the required N-i; Module D carries V1 rather than V2 certificates; twenty-three of twenty-four families have identity answer transport, so τ_T is unexercised; the surface controls are certified but not complexity-matched; and the evaluator audit is four families per module against the ninety-four that (56) requires. What the bank does instantiate—schema, certificates, transport laws, anti-certificates, admission invariants, and the analysis path end to end—is listed below. The modules are: A quotient and gauge; B local-to-global; C analytic legality; D anomaly and measure; E boundary; F representation and

naturality transport (Section 8.6). Each family is recorded under a fixed template—canonical presentation, two certified structural transforms with transported reference answers, composite reference answer, certified surface control, anti-certified negative control, answer type, certificate facets, and executable certificate digest—under three replayability rules: typed data rather than prose is authoritative; every convention (signature, orientation, normalization, basis) lives in the typed data and never as an unstated default; and reference values serialize exactly, with integers as integers and rationals as p/q, never as floating-point.

Verification status, stated exactly. The shipped verification script checks every encoded reference value, and for the twenty V2 families—modules A, B, C, E, F—it *independently re-derives* that value symbolically in a computer-algebra system before admission. For the four Module D families it performs a consistency check under a declared literature convention: the normalization is transcribed rather than derived, and what the script verifies is algebraic consistency of the ledger, not the physics. Those four are excluded from the confirmatory estimand until promoted (threat T4). An earlier version of this work described both operations as symbolic re-derivation; that conflation is corrected here.

Corrections carried into the bank. Two are substantive. First, the Module D family concerning the index records the *index of the chiral operator* $\mathrm{ind}\not{D}_+) = 2T(R)k = 1$ for the fundamental of $SU(2)$ at $k = 1$; the integrated axial-charge violation for a Dirac fermion is $\Delta Q_5 = 2\,\mathrm{ind}\not{D}_+) = 2$ by (39), and any item stem asking for "the integrated anomaly" must ask for the latter. An earlier statement identifying the two was wrong by a factor of two and is corrected in Section 8.3. Second, the bank's negative controls are predominantly of type N-ii (false-claim controls) in the sense of Definition 6.15: valuable for the obstruction track, but not admissible in the confirmatory anti-equivalence role, for which type N-i neighbouring-problem controls are required. Supplying N-i controls for every family is a pre-run deliverable.

Released artifacts, with an exact scope table. The released software is a *primary paired-endpoint and inference-engine prototype*, not an implementation of Section 9 as a whole, and the following table states which is which.

Section 9 component	Released	Where
Paired observation record, JP(T), δ_T	yes	finite transports verified extensionally; production typed transports are package-owned symbolic maps whose executable implementations are generated internally and issued only through the sealed registry
Failure taxonomy: apparatus, refusal, parse, answer-type, canonical, transport	yes	`observation.py`: failure router
Failure taxonomy: mechanism, witness	no	obligation labels; not derivable from a final-answer record
Metric suite $D^{\mu}, D^{t}, A_{\mathscr{G}}$ (arrow level); A_{orb} (family level over $\{\text{id}, T_1, T_2\}$); tie rates	yes	`observation.metric_suite`
Threshold curve and differential-evaluator sensitivity, on the deciding engine	yes	confirmatory result
N-i negative-control gate	yes	package-pinned registry, digest-bound, replayed internally, family-bound
Partial identification, decision rule, gates, fallback	yes	`analysis.py`: confirmatory entry point
Frozen admission, module universe, audit provenance	yes	`admission.json`, `analysis`
Obligation transport λ_T, δ_T^P	prototype	`proof_summary.py`, not wired into the primary path
Latent-annotator (Dawid–Skene) aggregation	no	analysis plan only
Live evaluator audit of renderer/parser/judge	no	analysis plan only
H2, H3	no	descriptive / design target
Generation harness	no	no model is run in this work

The bundle contains the measurement core; the family definitions; the materializer; the dry audit; the manifest generator; the parser alias table; the paired-observation module; the analysis prototype; the proof-summary module implementing λ_T, its three laws, and δ_T^P with its declared edge cases—an *isolated prototype*, unit-tested but not wired into the primary response path, the gates, or the family schema, because the released bank is final-answer only; the frozen admission manifest `admission.json`; the verbatim system and format-repair prompts; the verification script; the test suite; six symbolic certificate scripts; and the twenty-four materialized family folders. The bank itself contains *final-answer* families only, so the proof-summary layer is exercised by unit tests rather than by the bank.

Four operational layers are *not* released and are not claimed to be: the latent-annotator (Dawid–Skene) aggregation of human obligation records; a live evaluator audit of a renderer, parser, human grader or LLM judge—what `core.py` computes is kernel/gold self-consistency, which compares frozen reference objects through the same equality kernel that defined them and is named accordingly; the H2 and H3 comparisons; and the generation harness itself, since no model is run anywhere in this work. In the environment described in the accompanying build notes and in `anc/RUNNING.md`, executed from a fresh extraction with no manual step: the verification script re-derives the twenty V2 reference values symbolically and checks the four Module-D ledger values; the dry audit passes all eight admission invariants for all twenty-four families; the analysis prototype runs on synthetic replicate-level records and reports its gates in the pre-specified order; the test suite reports 302 passing tests, including release-integrity tests that recompute every digest and assert that the manifest entry set equals the released file set minus itself; and the manifest pins 105 files by SHA-256. Three hashes are quoted so that the bundle can be checked without trusting it:

```
core.py          0abaa20f2a7aec22844ad6b4a893a1e8c00ae87a5dac24b46b55b02f4f1e8ef6
analysis.py      1feeb58ef6864fa67e9732f23f019b113e3ac2f5342abe3ec1f6273f69f3b591
MANIFEST.sha256 7e54d898ff1a954adb4a43393ff59f9cbe50bdb34f3743058d73f2e752c209a4
```

The third line is the root of trust: `MANIFEST.sha256` pins the other 105 files and does not pin itself, so verifying it verifies the bundle. Decision constants fixed by Section 9: $\delta_{\text{sem}} = 0.05$; evaluator gate 0.05 on the one-sided upper bound; four distinct bootstrap counts $B_{\text{prim}} = 20{,}000$, $B_{\text{wild}} = 4{,}000$, $B_{\text{curve}} = 4{,}000$, $B_{\text{sens}} = 2{,}000$; master seed 20240517.

Outstanding items before any confirmatory run. Non-identity-transport families sufficient for the stratum estimand (T2); complexity-matched surface controls with the reported matching statistics (T3); promotion of the Module D certificates (T4); N-i negative controls for confirmatory use; renderer-and-judge audit on expert-equivalent artifacts rather than rendered reference values; extensional testing of independently specified composite transports, the current tests verifying composites that are defined as composites; an evaluator audit of at least ninety-four *families* per module, per (56), without which the simultaneous evaluator gate of Section 9 cannot be met at all; and public deposit of the protocol. Until these are complete, the instrument is a released design, not a validated measuring device, and this paper describes it as such.

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Index

answer transport τ_T, 17
anti-equivalence certificate, 23
artifact collapse, 9
artifact non-identifiability, 9
artifact projection, 9

benchmark six-tuple, 7

certificate ladder V0–V4, 23
cluster bootstrap, 37
construct validity, 3
correctness subfunctor, 17

decision rule, 34

evaluator presentation-consistency defect D_{eval}, 40

family **F**, 29
finite-sample non-identification, 9
formalization adequacy, 11

groupoid $\mathscr{G}_{\text{IMR}}$, 16

holonomy condition, 18

identification problem, 8
invariance defect $D_{\mathscr{G}}$, 21

obligation vector **J**, 32

proof summary, 19

quotient non-identification, 22
quotient validity, 20

semantic normalizer Π_x, 19
structural-arm excess Δ_{paired}, 34
surface subgroupoid $\mathscr{G}_{\text{surf}}$, 30

transport equivariance, 21
transport-stable correctness, 25

verifier-selection bias, 10